

A Qubit in an Oscillator:

From Ideal GKP Codes to Energy-Scaled Error
Mitigation

by

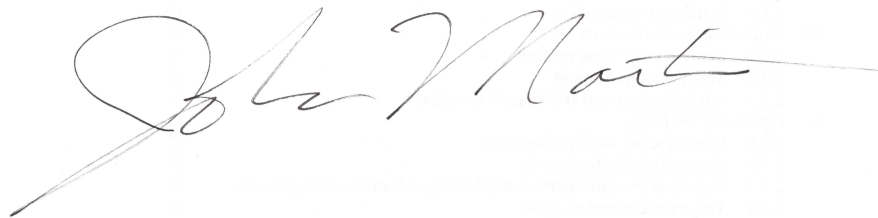
Guizhong (Harry) Luo

A thesis submitted in partial fulfillment
of the requirements for the degree of
Bachelor of Science
(Physics Honors)

at the
University of Wisconsin-Madison
Spring, 2026

A Qubit in an Oscillator:
From Ideal GKP Codes to Energy-Scaled Error
Mitigation

by
Guizhong (Harry) Luo

A handwritten signature in black ink, appearing to read "John Martinis", is written across the center of the page. The signature is fluid and cursive, with a long horizontal stroke extending to the right.

A thesis submitted in partial fulfillment
of the requirements for the degree of
Bachelor of Science
(Physics Honors)

at the
University of Wisconsin-Madison
Spring, 2026

Photocopy of the thesis frontpage, signed by John M. Martinis, 2025 Nobel laureate in Physics,
during his visit to Madison on Mar. 23, 2026.

Contents

1	A Qubit from Displacements: The Ideal GKP Code	4
1.1	The Harmonic Oscillator	4
1.2	Displacement Operations in Phase Space	5
1.2.1	Composition and the area condition	6
1.3	The Square-Lattice GKP Code	8
1.3.1	Codewords	9
1.3.2	Logical operators	10
1.3.3	Correctable region	11
1.4	The Wigner Function	12
1.4.1	Wigner function of ideal GKP codewords	13
2	Physical Realization of the GKP Code	16
2.1	The Physical Setting	16
2.1.1	From harmonic cavity to anharmonic ancilla	16
2.1.2	Dispersive coupling and conditional control	18
2.1.3	Hardware parameters	19
2.2	Finite-Energy Codewords	20
2.2.1	Finite-energy construction	20
2.2.2	Mean photon number	22
2.2.3	Intrinsic error and the Wigner function	22
2.3	Operating the Code	24
2.3.1	Logical gates as displacements	24
2.3.2	Controlled displacements	24
2.3.3	Stabilizer measurement	25
2.3.4	The error-correction cycle	26
2.3.5	Fast conditional displacements and state preparation	26
2.4	Pure Photon Loss During the QEC Cycle	27
2.4.1	From closed to open evolution	28
2.4.2	How loss distorts the GKP grid	30
2.5	Choosing the Operating Point	31
3	Recovery of GKP States Under Loss	33
3.1	The Logical Channel Framework	33
3.1.1	Orthonormalization	33
3.1.2	Encoding and decoding maps	34
3.2	Knill-Type Error Correction	35
3.3	Syndromes Decoding	36
3.4	The Petz Recovery Map	36
3.5	Near-Optimal Fidelity	38
3.6	The Decoded Logical Map	39

4	Energy-Scaled Zero-Noise Extrapolation	42
4.1	Quantum Error Mitigation and Zero-Noise Extrapolation	42
4.2	The Extrapolation Target and Energy Schedule	43
4.3	The ES-ZNE Protocol	45
4.4	Single-Qubit Results and Regime Analysis	48
4.5	Two-Qubit Extension and Resource Analysis	50
5	Beyond the Thesis: Open Questions and Future Directions	54
5.1	Beyond Pure Photon Loss	54
5.2	Beyond the Square-Lattice Qubit Code	56
5.3	Beyond GKP, and the Monotonicity Premise of ES-ZNE	57

ABSTRACT

Bosonic error-correcting codes encode quantum information in oscillator modes and offer a route toward hardware-efficient quantum computing. The Gottesman-Kitaev-Preskill (GKP) code is a central example, but its ideal states require infinite energy, while realistic implementations must operate with finite squeezing and noise. This thesis develops the GKP code from its ideal phase-space construction to its physical realization in superconducting cavity-transmon systems. We then study finite-energy GKP states under pure photon loss and examine how their recovered logical performance changes as the code energy is varied. We find that increasing the code energy can improve performance over the sampled finite-energy window, and we use this behavior to propose an energy-scaled zero-noise extrapolation method that is compatible with current GKP platforms. We close by discussing extensions to other noise models, lattice choices, and bosonic codes.

PREFACE

This thesis intends to be a pedagogical reflection on the research I carried out as an undergraduate and the lessons that came with it. Its central project is based primarily on a manuscript prepared over the past year with Prof. Matthew Otten, currently under review. The technical question is how ideal Gottesman-Kitaev-Preskill codes, which encode a qubit into an oscillator, behave when they meet the finite energy and photon loss of real physical systems. The personal question, which I understood only gradually, was how one learns to do research when the next step is no longer assigned.

I have always been drawn to physics as a way of seeking an isomorphism between physical reality and theoretical abstraction. In this project I had the opportunity to find such a mapping locally. The ideal GKP code is clean, algebraic, and geometric; the physical code is approximate, noisy, and constrained. Much of this thesis lives in the space between those two descriptions. Over the past twenty months I learned a great deal of physics, mathematics, simulation, literature, and writing. The steepest part was independence. The hardest lesson was not any single derivation or program, but learning to decide what question was worth asking next, and then taking responsibility for the answer.

I owe my gratitude to Prof. Matt Otten, who trusted me with that independence before I fully knew what to do with it. His mentorship made this work possible, by giving me room to think, fail, revise, and return with something of my own. However, any errors which remain are my sole responsibility. I am also grateful to Prof. Ilya Esterlis, who first showed me what real physics is about; to Prof. Alex Levchenko, who removed the friendly sugarcoating of physics and showed me the expectations placed on an aspiring researcher; to Prof. Lisa Everett and Prof. Elio Koenig, for candid discussions on quantum mechanics and life; and to Prof. Chris Rycroft, who gave me my first opportunity to join analysis with mechanics in the way I had long wanted.

Balancing research with a triple major is hard, and it would have been only a grind and no fun had it not been for the friends I made along the way. To this end I am particularly indebted to Yufei Shen for an intellectual companionship.

I also thank my family for their trust and encouragement, even when the details of the work were far from easy to explain.

As for Madison, I do not yet know how to write about it— not without a distance. It is still the ordinary place where all of this happened. Perhaps I will understand it later, when what now feels ordinary has become memory, and maybe away from Madison could I write about Madison.

当时只道是寻常。

ACKNOWLEDGEMENT

I acknowledge the use of generative artificial intelligence tools, including ChatGPT, Claude, and Gemini, in the work that supported this thesis. These tools assisted with discussion, drafting, revision, code review, and organization at different stages of the project. Their rapid development marks a new phase of research practice, but their use also requires judgment, verification, responsibility, and taste. The arguments, calculations, and final presentation of this thesis remain my own.

This thesis and related works are supported by the NSF QLCI for Hybrid Quantum Architectures and Networks (NSF Award 2016136) and the UW-Madison L&S Honors Program through the Trewartha Senior Honors Thesis Research Grant.

SECTION 1

A Qubit from Displacements: The Ideal GKP Code

SUBSECTION 1.1

The Harmonic Oscillator

A qubit lives in a two-dimensional Hilbert space. An arbitrary unknown error cannot be detected or corrected, because two dimensions offer no room for redundancy. To protect quantum information, we embed it in a larger space. The approach of this thesis is to use a single mode of the electromagnetic field in a superconducting microwave cavity. Its Hilbert space is infinite-dimensional and therefore provides the redundancy needed for error correction [1].¹

A microwave cavity is an electromagnetic resonator. The field inside supports a discrete set of standing-wave modes, each characterized by a resonant frequency ω_m . In the superconducting cavities used for GKP experiments, one such mode can be isolated well enough that it behaves as a single dynamical degree of freedom [3]. In circuit language, the electric and magnetic energy of that mode are written as $Q^2/2C$ and $\Phi^2/2L$, where Q and Φ are the charge and flux associated with the mode, and C and L are the effective capacitance and inductance [3]. The total energy is

$$H_{\text{LC}} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}. \quad (1.1)$$

This is a harmonic oscillator because it has the standard oscillator form $p^2/2m + m\omega^2 x^2/2$, with the identifications $m \leftrightarrow C$, $x \leftrightarrow \Phi$, $p \leftrightarrow Q$, and $\omega_r = 1/\sqrt{LC}$.

Equation (1.1) is still classical. To pass to the quantum description, we quantize this single mode in the usual canonical way: elevating the classical variables Φ and Q to operators $\hat{\Phi}$ and $\hat{Q}$ satisfying $[\hat{\Phi}, \hat{Q}] = i\hbar$. It is convenient to introduce the annihilation operator (also known as the ladder operator, named through its role in raising and lowering energy levels) defined as

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega_r C}}(C\omega_r \hat{\Phi} + i\hat{Q}),$$

for which $[\hat{a}, \hat{a}^\dagger] = 1$. Substituting into Eq. (1.1) gives

$$\hat{H} = \hbar\omega_r \left(\hat{n} + \frac{1}{2} \right), \quad (1.2)$$

where $\hat{n} = \hat{a}^\dagger \hat{a}$ is the photon number operator. A single cavity mode is therefore a bosonic quantum harmonic oscillator, and $\hat{a}^\dagger$ creates one photon in that mode. The energy eigenstates are the Fock states $\{|n\rangle\}_{n=0}^\infty$, satisfying $\hat{n}|n\rangle = n|n\rangle$. They form an orthonormal basis for the oscillator Hilbert space $\mathcal{H} = \text{span}\{|n\rangle\}_{n=0}^\infty$. The ladder operators act on this basis as

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (1.3)$$

¹ In the experiments of Campagne-Ibarcq et al. [2], this mode is a reentrant coaxial microwave cavity with photon lifetime $T_1 = 245 \mu\text{s}$ and mean photon number $\bar{n} \approx 8$ –10.

The number basis is natural for energy, but the GKP code will be built from translations in phase space. For that reason, we introduce two Hermitian combinations of $\hat{a}$ and $\hat{a}^\dagger$, the quadrature operators:²

$$\hat{q} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{p} = \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}}. \quad (1.4)$$

² These dimensionless quadrature conventions are standard in the continuous-variable quantum information literature [4, 5].

These are the dimensionless versions of flux and charge,

$$\hat{q} = \frac{\hat{\Phi}}{\sqrt{\hbar Z_r}}, \quad \hat{p} = \sqrt{\frac{Z_r}{\hbar}} \hat{Q}, \quad Z_r = \sqrt{L/C}.$$

Using $[\hat{\Phi}, \hat{Q}] = i\hbar$, we find

$$[\hat{q}, \hat{p}] = i. \quad (1.5)$$

These play the role of dimensionless position and momentum for the oscillator. The pair $(\hat{q}, \hat{p})$ defines a two-dimensional phase space, a plane in which each point specifies the mean values of both quadratures. We make this phase-space picture precise in §1.4 with the Wigner function.

We set $\hbar = 1$ and absorb the frequency into the definition of unit-time, so that

$$\hat{H} = \hat{n} + \frac{1}{2}. \quad (1.6)$$

Error correction takes place in the infinite-dimensional oscillator Hilbert space $\mathcal{H}$, which provides room to embed a protected two-dimensional subspace. To construct such a subspace, we need to identify operations on the oscillator that can serve as building blocks. The natural candidates are translations in phase space.

SUBSECTION 1.2

Displacement Operations in Phase Space

The simplest operations in phase space are rigid shifts along one quadrature.

Recall that the momentum operator $\hat{p}$ generates translations in position: acting on a position eigenstate,

$$e^{-iu\hat{p}} |q\rangle = |q + u\rangle. \quad (1.7)$$

The unitary $e^{-iu\hat{p}}$ shifts every position eigenstate by $+u$ while leaving the momentum distribution unchanged.³ Similarly, $\hat{q}$ generates translations in momentum:

$$e^{iv\hat{q}} |p\rangle = |p + v\rangle. \quad (1.8)$$

³ Equation (1.7) follows from $[\hat{q}, \hat{p}] = i$ via the identity $e^{-iu\hat{p}} \hat{q} e^{iu\hat{p}} = \hat{q} + u$.

For a shift by an arbitrary phase-space vector $(\Delta q, \Delta p)$, we combine both generators. Parametrizing by $\alpha = (\Delta q + i\Delta p)/\sqrt{2} \in \mathbb{C}$, we define:

Definition 1 (Displacement operator) The displacement operator for $\alpha \in \mathbb{C}$ is the unitary

$$\hat{D}(\alpha) = \exp(\alpha \hat{a}^\dagger - \alpha^* \hat{a}). \quad (1.9)$$

Its action on a state is a rigid translation in phase space by the vector $(\sqrt{2} \operatorname{Re} \alpha, \sqrt{2} \operatorname{Im} \alpha)$.

To verify this, we express the exponent in terms of the quadratures (1.4). Writing $\alpha = \alpha_r + i\alpha_i$ with $\alpha_r, \alpha_i \in \mathbb{R}$,

$$\alpha \hat{a}^\dagger - \alpha^* \hat{a} = -i\sqrt{2} \alpha_r \hat{p} + i\sqrt{2} \alpha_i \hat{q}.$$

For a real displacement ($\alpha_i = 0$),

$$\hat{D}(\alpha) = e^{-i\alpha\sqrt{2}\hat{p}}, \quad (1.10)$$

which is the position-translation (1.7) with $u = \alpha\sqrt{2}$. For a purely imaginary displacement $\alpha = i\beta$ with $\beta \in \mathbb{R}$,

$$\hat{D}(i\beta) = e^{i\beta\sqrt{2}\hat{q}}, \quad (1.11)$$

which is the momentum-translation (1.8) with $v = \beta\sqrt{2}$. The general displacement combines both shifts with a global phase.

1.2.1 Composition and the area condition

Two displacements composed in sequence do not simply produce the displacement by the sum of their arguments, because the noncommutativity (1.5) introduces a phase. To see why, write $\hat{A} = \alpha \hat{a}^\dagger - \alpha^* \hat{a}$ and $\hat{B} = \beta \hat{a}^\dagger - \beta^* \hat{a}$, so that $\hat{D}(\alpha) = e^{\hat{A}}$ and $\hat{D}(\beta) = e^{\hat{B}}$. Their commutator is

$$[\hat{A}, \hat{B}] = \alpha\beta^* - \alpha^*\beta,$$

a *c*-number.⁴ When the commutator of two operators is a *c*-number, the Baker–Campbell–Hausdorff (BCH) formula simplifies. Recall that in general,

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A} + \hat{B} + \frac{1}{2}[\hat{A}, \hat{B}] + \dots},$$

where the higher-order terms involve nested commutators $[\hat{A}, [\hat{A}, \hat{B}]]$, $[\hat{B}, [\hat{A}, \hat{B}]]$, and so on. But if $[\hat{A}, \hat{B}]$ is a scalar, all nested commutators vanish, and the series terminates:

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A} + \hat{B} + [\hat{A}, \hat{B}]/2}. \quad (1.12)$$

Applying this to the two displacement operators, the composition law is

$$\hat{D}(\alpha) \hat{D}(\beta) = e^{(\alpha\beta^* - \alpha^*\beta)/2} \hat{D}(\alpha + \beta). \quad (1.13)$$

Swapping $\alpha \leftrightarrow \beta$ in (1.13) gives

$$\hat{D}(\beta) \hat{D}(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2} \hat{D}(\alpha + \beta),$$

⁴ A *c*-number (classical number) is a scalar that commutes with every operator. The term originates from Dirac's distinction between *c*-numbers and *q*-numbers (operator-valued quantities).

with the opposite sign in the exponent. Both expressions share the factor $\hat{D}(\alpha+\beta)$ on the right-hand side. Solving for it in each and equating,

$$\hat{D}(\alpha) \hat{D}(\beta) = e^{\alpha\beta^* - \alpha^*\beta} \hat{D}(\beta) \hat{D}(\alpha). \quad (1.14)$$

The phase factor $\alpha\beta^* - \alpha^*\beta$ in (1.14) has a geometric interpretation. It measures a signed area in phase space. Writing $\alpha\beta^* - \alpha^*\beta = 2i \operatorname{Im}(\alpha\beta^*)$, we translate the complex-number expression into the language of phase-space vectors.⁵ Let

$$\mathbf{u} = (\sqrt{2} \operatorname{Re} \alpha, \sqrt{2} \operatorname{Im} \alpha), \quad \mathbf{v} = (\sqrt{2} \operatorname{Re} \beta, \sqrt{2} \operatorname{Im} \beta)$$

be the phase-space displacement vectors. Then

$$\operatorname{Im}(\alpha\beta^*) = \frac{1}{2}(u_p v_q - u_q v_p) = -\frac{1}{2} \mathbf{u}^T \Omega \mathbf{v},$$

where

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (1.15)$$

is the symplectic form.⁶ The quantity $\mathbf{u}^T \Omega \mathbf{v}$ is the *signed area* of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$ in phase space. Substituting into (1.14),

$$\hat{D}(\alpha) \hat{D}(\beta) = e^{-i \mathbf{u}^T \Omega \mathbf{v}} \hat{D}(\beta) \hat{D}(\alpha). \quad (1.16)$$

Two displacements commute when this phase equals unity.

Theorem 1

(Area condition) Two displacement operators $\hat{D}(\alpha)$ and $\hat{D}(\beta)$ commute if and only if the signed area of the parallelogram spanned by their phase-space vectors is an integer multiple of 2π ,

$$\hat{D}(\alpha) \hat{D}(\beta) = \hat{D}(\beta) \hat{D}(\alpha) \iff \mathbf{u}^T \Omega \mathbf{v} \in 2\pi\mathbb{Z}. \quad (1.17)$$

This is the central geometric condition underlying the GKP code [1, 6]. In physical terms, two displacements commute precisely when the parallelogram they span encloses an integer number of *Planck cells*, each of area 2π (set by $[\hat{q}, \hat{p}] = i$).

The area condition tells us when an entire lattice of phase-space translations can be imposed consistently, making an invariant encoded subspace possible. The idea is to make certain large displacements act trivially on the encoded states. If a state is unchanged by translating it by one of the lattice vectors, then the encoded information no longer depends on its absolute location in phase space, but only on where it sits within a fundamental cell. That is the kind of redundancy an error-correcting code needs.

A set of displacements whose pairwise signed areas are all multiples of 2π forms such a commuting family. Because they commute, we can consistently require a state to have a definite eigenvalue under all of them at once. Choosing the common +1 eigenspace means selecting the subspace of states that are

⁵ The role of unit-cell area in the commutation structure originates in Gottesman, Kitaev, and Preskill [1]. Conrad [6] develops this into an explicit geometric viewpoint, connecting the phase-space parallelogram to the quantum Hall Hamiltonian.

⁶ The symplectic form Ω is the natural area measure on two-dimensional phase space. For two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, the product $\mathbf{u}^T \Omega \mathbf{v} = u_q v_p - u_p v_q$ equals the signed area of the parallelogram they span—positive for counter-clockwise orientation, negative for clockwise. It plays the same role as the cross product in three-dimensional geometry, but lives in even-dimensional spaces. Symplectic geometry is the mathematical framework for Hamiltonian mechanics; here we need only the area interpretation.

invariant under the chosen lattice translations:

$$\hat{U}_k |\psi\rangle = |\psi\rangle \quad \text{for every generator } \hat{U}_k.$$

This invariant subspace is the *code space*. Other common eigenspaces carry the same lattice structure but with different phase labels, so the +1 sector is a convenient reference choice.

The commuting displacements that fix the code space are called *stabilizers*. The term is borrowed from group theory, where the stabilizer of a point under a group action is the subgroup that fixes it. Here, each stabilizer leaves every state in the code space invariant. We now construct the simplest example.

SUBSECTION 1.3

The Square-Lattice GKP Code

With the area condition and the stabilizer concept from the preceding section in hand, we now construct the square-lattice GKP code. The simplest tiling of phase space by a lattice of commuting displacements is a square lattice. We choose generators that displace by $2\sqrt{\pi}$ along each quadrature axis:

Definition 2

(Square-lattice GKP code) The **square-lattice GKP code** is defined by two stabilizer generators,

$$S_q = \hat{D}(\sqrt{2\pi}) = e^{-i2\sqrt{\pi}\hat{p}}, \quad S_p = \hat{D}(i\sqrt{2\pi}) = e^{+i2\sqrt{\pi}\hat{q}}, \quad (1.18)$$

which translate $\hat{q}$ and $\hat{p}$ by $+2\sqrt{\pi}$ respectively.

The **ideal code space** C is the formal simultaneous +1 eigenspace of the two lattice translations:

$$C = \{ |\psi\rangle : S_q |\psi\rangle = |\psi\rangle \quad \text{and} \quad S_p |\psi\rangle = |\psi\rangle \}. \quad (1.19)$$

The codespace C is therefore an ideal algebraic codespace, not yet a physical subspace of $\mathcal{H}$. Its exact codewords satisfy the stabilizer and logical-operator relations we want, but their wavefunctions are Dirac-comb distributions and cannot be normalized. In this sense the construction is formal: it defines the code by its algebraic structure first. Chapter 2 then replaces these ideal combs by finite-energy states that approximate the same lattice structure.

The two generators, S_p and S_q , commute. Using (1.10) and (1.11), the phase-space displacement vectors are $\mathbf{u} = (2\sqrt{\pi}, 0)$ for S_q and $\mathbf{v} = (0, 2\sqrt{\pi})$ for S_p , so the signed area is

$$\mathbf{u}^T \Omega \mathbf{v} = 2\sqrt{\pi} \cdot 2\sqrt{\pi} = 4\pi = 2 \times 2\pi.$$

By the area condition (1.17), S_q and S_p commute. To determine what information this code protects, we need the dimension of C . We now show that the two stabilizer conditions together admit two linearly independent solutions, making C two-dimensional and encoding one logical qubit.

1.3.1 Codewords

The states satisfying $S_q |\psi\rangle = |\psi\rangle$ and $S_p |\psi\rangle = |\psi\rangle$ are the *codewords* of the code, and their span is the code space $\mathcal{C}$ (1.19). We solve for them.

The S_q condition. The stabilizer $S_q = e^{-i2\sqrt{\pi}\hat{p}}$ translates position eigenstates by $+2\sqrt{\pi}$,

$$S_q |q\rangle = |q + 2\sqrt{\pi}\rangle.$$

Projecting the eigenvalue condition $S_q |\psi\rangle = |\psi\rangle$ onto $\langle q|$ gives

$$\langle q|S_q|\psi\rangle = \langle q|\psi\rangle \implies \langle q - 2\sqrt{\pi}|\psi\rangle = \langle q|\psi\rangle,$$

where the sign flip follows from $S_q^\dagger |q\rangle = |q - 2\sqrt{\pi}\rangle$. Thus the position wave function is periodic with period $2\sqrt{\pi}$. By itself this is a weak constraint: any $2\sqrt{\pi}$ -periodic distribution satisfies the S_q condition.

The S_p condition. The stabilizer $S_p = e^{i2\sqrt{\pi}\hat{q}}$ acts diagonally in the position basis,

$$S_p |q\rangle = e^{i2\sqrt{\pi}q} |q\rangle.$$

Projecting the eigenvalue condition $S_p |\psi\rangle = |\psi\rangle$ onto $\langle q|$ gives

$$\langle q|S_p|\psi\rangle = \langle q|\psi\rangle \implies e^{i2\sqrt{\pi}q} \psi(q) = \psi(q).$$

Hence either $\psi(q) = 0$ or $e^{i2\sqrt{\pi}q} = 1$, which requires $q \in \sqrt{\pi}\mathbb{Z}$. The wave function is supported on the fine lattice $\Lambda = \sqrt{\pi}\mathbb{Z}$, whose spacing is half the stabilizer period. By itself this is again a weak constraint: arbitrary weights on that lattice still satisfy the S_p condition.

Two codewords. Combining both conditions, the wave function must be $2\sqrt{\pi}$ -periodic (from S_q) and supported only on $\Lambda = \sqrt{\pi}\mathbb{Z}$ (from S_p). The allowed support points are therefore fixed to the fine lattice, while the periodicity identifies points separated by $2\sqrt{\pi}$ and forces their amplitudes to agree. The only remaining freedom is the choice of an initial offset on Λ , so the wave function must be a Dirac comb with that period:

$$|\psi\rangle \propto \sum_{s=-\infty}^{\infty} |q = q_0 + 2s\sqrt{\pi}\rangle, \quad (1.20)$$

where $q_0 \in \Lambda$ is an offset yet to be fixed (the periodicity maps peak s to peak $s + 1$, requiring equal amplitudes). The offset must lie on Λ , but the S_q periodicity identifies offsets that differ by $2\sqrt{\pi}$. This identification defines a coarser sublattice⁷ $\Lambda_S = 2\sqrt{\pi}\mathbb{Z} \subset \Lambda$. Since Λ_S is a subgroup of Λ under addition, the allowed offsets partition into cosets of Λ_S :

$$\begin{aligned} \Lambda_S + 0 &= \{\dots, -2\sqrt{\pi}, 0, 2\sqrt{\pi}, \dots\}, \\ \Lambda_S + \sqrt{\pi} &= \{\dots, -\sqrt{\pi}, \sqrt{\pi}, 3\sqrt{\pi}, \dots\}. \end{aligned}$$

⁷ Given a group G and a subgroup $H \leq G$, a coset of H is a set $H + g = \{h + g : h \in H\}$ for some $g \in G$. The distinct cosets partition G into non-overlapping classes of equal size. Their number is the index $[G : H]$. Lagrange's theorem states

$$|G| = [G : H] \cdot |H|$$

for finite groups; for infinite lattices like $\mathbb{Z}$, the index equals the ratio of generator spacings.

These two cosets exhaust Λ with no overlap. The quotient group and its index are

$$\Lambda / \Lambda_S = \{\Lambda_S + 0, \Lambda_S + \sqrt{\pi}\} \cong \mathbb{Z}_2,$$

$$[\Lambda : \Lambda_S] = \frac{2\sqrt{\pi}}{\sqrt{\pi}} = 2.$$

Each coset gives one codeword, a Dirac comb with spacing $2\sqrt{\pi}$ offset by 0 or $\sqrt{\pi}$:

$$|0_L\rangle \propto \sum_{s=-\infty}^{\infty} |q = 2s\sqrt{\pi}\rangle, \quad |1_L\rangle \propto \sum_{s=-\infty}^{\infty} |q = (2s+1)\sqrt{\pi}\rangle. \quad (1.21)$$

Since the two cosets are disjoint, the codewords are orthogonal:

$$\langle 0_L | 1_L \rangle \propto \sum_{s,s'} \delta(2s\sqrt{\pi} - (2s'+1)\sqrt{\pi}) = 0.$$

The code space is two-dimensional, encoding one logical qubit [1, 7], and the two basis states differ by a displacement of $\sqrt{\pi}$ in position, exactly half the stabilizer period.⁸

1.3.2 Logical operators

Having defined the code space via its stabilizers and identified the two basis states, we need operators that act as Pauli gates on the encoded qubit. These are the *logical operators*. A logical operator must preserve the code space (commuting with every stabilizer) but act nontrivially within it, distinguishing the two codewords. By contrast, stabilizers also preserve the code space but act *trivially*: every codeword is a +1 eigenstate, so the stabilizer does nothing to encoded information.

Since $|0_L\rangle$ and $|1_L\rangle$ differ by a shift of $\sqrt{\pi}$ in position, the operator that maps $|0_L\rangle \leftrightarrow |1_L\rangle$ is a displacement by $\sqrt{\pi}$ along $\hat{q}$. Applying this operator twice shifts by $2\sqrt{\pi}$, recovering the stabilizer S_q , so the logical $\bar{X}$ is a displacement by *half* the stabilizer period. Similarly, the operator that applies a relative phase distinguishing the two codewords is a displacement by $\sqrt{\pi}$ along the momentum axis, giving the logical $\bar{Z}$. Explicitly:

$$\bar{X} = \hat{D}\left(\sqrt{\frac{\pi}{2}}\right) = e^{-i\sqrt{\pi}\hat{p}}, \quad \bar{Z} = \hat{D}\left(i\sqrt{\frac{\pi}{2}}\right) = e^{+i\sqrt{\pi}\hat{q}}. \quad (1.22)$$

We now compute the action of $\bar{X}$. It shifts the position quadrature by $+\sqrt{\pi}$.

Here $\propto$ indicates a formal superposition: the ideal codewords are non-normalizable, with $\langle q | q' \rangle = \delta(q - q')$ and infinite mean photon number. Chapter 2 introduces finite-energy approximations.

⁸ This generalizes: for a rectangular GKP lattice with stabilizer spacing $d\sqrt{\pi}$, the index becomes $[\sqrt{\pi}\mathbb{Z} : d\sqrt{\pi}\mathbb{Z}] = d$, and the code encodes a d -dimensional qudit. The square-lattice qubit code is the $d = 2$ case.

Indeed:

$$\bar{X} |0_L\rangle \propto \sum_s \bar{X} |2s\sqrt{\pi}\rangle \quad (1.23)$$

$$= \sum_s |2s\sqrt{\pi} + \sqrt{\pi}\rangle \quad (1.24)$$

$$= \sum_s |(2s+1)\sqrt{\pi}\rangle \quad (1.25)$$

$$= |1_L\rangle, \quad (1.26)$$

and $\bar{X} |1_L\rangle = |0_L\rangle$ by the same argument.

For $\bar{Z}$, the action on a position eigenstate is a phase:

$$\bar{Z} |q\rangle = e^{i\sqrt{\pi}q} |q\rangle.$$

On the codeword peaks at $q = m\sqrt{\pi}$ (integer m), this phase evaluates to

$$e^{i\sqrt{\pi} \cdot m\sqrt{\pi}} = e^{i\pi m} = (-1)^m.$$

For $|0_L\rangle$, all peaks sit at even m , so $(-1)^m = +1$ and

$$\bar{Z} |0_L\rangle = + |0_L\rangle.$$

For $|1_L\rangle$, all peaks sit at odd m , so $(-1)^m = -1$ and

$$\bar{Z} |1_L\rangle = - |1_L\rangle.$$

The anticommutation $\bar{X}\bar{Z} = -\bar{Z}\bar{X}$ follows from the area condition (1.17). The displacement vectors for $\bar{X}$ and $\bar{Z}$ are $\mathbf{u} = (\sqrt{\pi}, 0)$ and $\mathbf{v} = (0, \sqrt{\pi})$, spanning a parallelogram of signed area $\mathbf{u}^T \Omega \mathbf{v} = \pi$. By (1.16), the phase is $e^{-i\pi} = -1$:

$$\bar{X}\bar{Z} = -\bar{Z}\bar{X}. \quad (1.27)$$

This is the Pauli algebra for a single qubit, emerging from the geometry of phase space [1, 5]. Notice the contrast with the stabilizers, where the area was 4π and the phase $e^{-i4\pi} = +1$. The logical operators live at half the lattice period, and it is this half-period structure that produces the anticommutation. Both stabilizers and logical operators are displacement operators, elements of the same group. The distinction is geometric. Stabilizers correspond to the full lattice vectors (cell area 4π , phase $e^{-i4\pi} = +1$, trivial action on the code space), while logical operators correspond to the half-lattice vectors (cell area π , phase $e^{-i\pi} = -1$, nontrivial action).

1.3.3 Correctable region

An error $\hat{D}(\epsilon)$ displaces the state by a small vector $(\Delta q, \Delta p) = (\sqrt{2} \operatorname{Re} \epsilon, \sqrt{2} \operatorname{Im} \epsilon)$ in phase space. The stabilizer measurement identifies this displacement only modulo the fine lattice spacing $\sqrt{\pi}$. If the displacement lies in the central decision cell, the syndrome identifies the nearest lattice representative unam-

biguously and the opposite displacement reverses the error. This nearest-lattice-point decoding succeeds when

$$|\Delta q| < \frac{\sqrt{\pi}}{2}, \quad |\Delta p| < \frac{\sqrt{\pi}}{2}. \quad (1.28)$$

The correctable region is therefore the square $[-\sqrt{\pi}/2, \sqrt{\pi}/2]^2$ centered at zero displacement [1]. Errors outside this cell are no longer uniquely distinguished by the syndrome. Rounding such an error back to the central cell can leave a residual displacement together with a logical half-lattice translation, and that logical component corrupts the encoded information.⁹

The square lattice is not the only valid choice. Any lattice Λ in phase space such that $\mathbf{u}^T \Omega \mathbf{v} \in 2\pi\mathbb{Z}$ for all $\mathbf{u}, \mathbf{v} \in \Lambda$, called a *symplectically integral* lattice [5, 6], generates a valid GKP code. A hexagonal lattice with the same stabilizer density has a larger Voronoi cell (better circle packing) and is more robust against small isotropic displacement errors [1, 5]. We use the square lattice throughout this thesis for simplicity and consistency with the majority of the literature.

SUBSECTION 1.4

The Wigner Function

The full code structure (stabilizers, codewords, logical operators, correctable region) is entirely determined by a lattice in phase space. To see this structure, we need a representation of quantum states in the (q, p) plane.

Definition 3

(Wigner function) The Wigner function of a state $\hat{\rho}$ is the quasi-probability distribution

$$W(q, p) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx e^{ipx} \left\langle q + \frac{x}{2} \left| \hat{\rho} \right| q - \frac{x}{2} \right\rangle. \quad (1.29)$$

The Wigner function is always real-valued, but unlike a classical probability distribution it can take negative values [4]. Its marginals recover the standard quantum-mechanical probability distributions. Integrating over momentum gives the position distribution:

$$\int_{-\infty}^{\infty} dp W(q, p) = \frac{1}{2\pi} \int dx \left\langle q + \frac{x}{2} \left| \hat{\rho} \right| q - \frac{x}{2} \right\rangle \underbrace{\int dp e^{ipx}}_{2\pi \delta(x)} = \langle q | \hat{\rho} | q \rangle.$$

Similarly, integrating over position gives the momentum distribution:

$$\int_{-\infty}^{\infty} dq W(q, p) = \langle p | \hat{\rho} | p \rangle.$$

These identities make $W(q, p)$ a faithful phase-space representation of the full quantum state.

The Wigner function can take negative values, and the GKP codewords, as we now show, exhibit extensive negativity [8]. It is interesting to note that this

⁹ Two displacements are equivalent “modulo a stabilizer” if they differ by a stabilizer element. Since stabilizers act trivially on the code space, such displacements have the same effect on encoded information. The danger outside the central decision cell is that the nearest representative may differ from the original displacement by a logical translation, not just by a stabilizer.

negativity is a nonclassical phase-space feature, arising from interference in the grid-state superposition.¹⁰

1.4.1 Wigner function of ideal GKP codewords

We derive the Wigner function for the ideal codeword $|0_L\rangle$, whose position wave function is the Dirac comb $\psi(q') = \langle q'|0_L\rangle \propto \sum_s \delta(q' - 2s\sqrt{\pi})$. Substituting into (1.29) with $\hat{\rho} = |0_L\rangle\langle 0_L|$,

$$W(q, p) \propto \int dx e^{ipx} \sum_{s, s'} \delta(q + \frac{x}{2} - 2s\sqrt{\pi}) \delta(q - \frac{x}{2} - 2s'\sqrt{\pi}).$$

The second delta function fixes $x = 2(q - 2s'\sqrt{\pi})$, and the first then requires $q = (s + s')\sqrt{\pi}$. Writing $t = s + s'$ and $n = s - s'$, the integral over x collapses to a sum over integer pairs (t, n) , with q pinned to $t\sqrt{\pi}$ and $x = 2n\sqrt{\pi}$. The sum separates into a position constraint and a momentum series:

$$W(q, p) \propto \sum_t \delta(q - t\sqrt{\pi}) \sum_{n \equiv t \pmod{2}} e^{i2pn\sqrt{\pi}}.$$

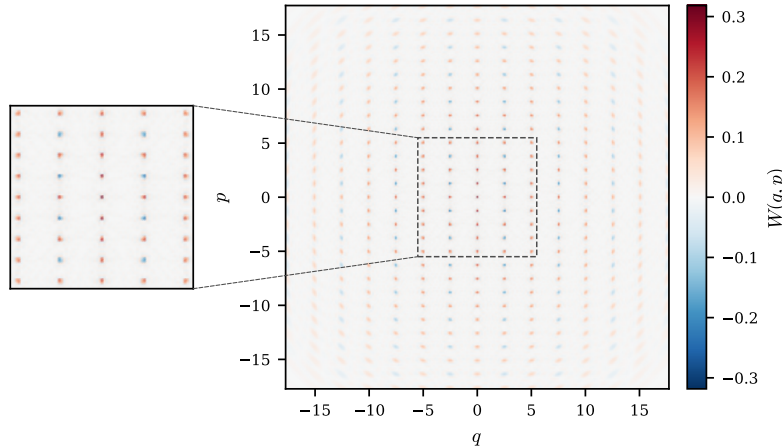


Figure 1. Wigner function of the ideal GKP codeword $|0_L\rangle$, evaluated numerically in a truncated Hilbert space. The grid spacing $2\sqrt{\pi}$ reflects the stabilizer period; the sublattice at $\sqrt{\pi}$ reflects the logical operators. Negative peaks (dark regions) form the checkerboard pattern of eq. (1.30), characteristic of GKP non-classicality.

The inner sum is evaluated by the Poisson summation identity $\sum_m e^{i\alpha m} = 2\pi \sum_k \delta(\alpha - 2\pi k)$, which converts it to a delta comb in p with spacing $\sqrt{\pi}/2$. When t is odd, the parity restriction $n \equiv 1 \pmod{2}$ introduces an extra phase $e^{i2p\sqrt{\pi}}$, which at each delta peak $p = s\sqrt{\pi}/2$ evaluates to $(-1)^s$. Combining both parities,

$$W^{(j)}(q, p) \propto \sum_{s, t=-\infty}^{\infty} (-1)^{s \cdot t} \delta\left(q - \sqrt{\pi} t - j\sqrt{\pi}\right) \delta\left(p - \frac{\sqrt{\pi}}{2} s\right), \quad (1.30)$$

¹⁰ Equivalently, the ideal GKP codewords are non-Gaussian. Gaussian states have nonnegative Gaussian Wigner functions, while the GKP grid state does not.

where $j = 0$ for $|0_L\rangle$ and $j = 1$ for $|1_L\rangle$ (the derivation for $|1_L\rangle$ is identical with the offset $q_0 = \sqrt{\pi}$). The peaks sit on a rectangular grid with spacing $\sqrt{\pi}$ in q (the logical operator period) and $\sqrt{\pi}/2$ in p [1]. The sign factor $(-1)^{s \cdot t}$, where the exponent is the product of the integer indices s and t , produces a checkerboard pattern: the sign is -1 only when both s and t are odd, and $+1$ otherwise. Three-quarters of the peaks are positive; one quarter are negative.

The coarse lattice spacing $2\sqrt{\pi}$ corresponds to the stabilizer period. The half-period sublattice at spacing $\sqrt{\pi}$ distinguishes $|0_L\rangle$ from $|1_L\rangle$ through the offset in q . The negative peaks are the Wigner negativity that serves as a nonclassical resource for quantum computation [8]. An illustration of the Wigner function of the GKP codeword $|0_L\rangle$ is shown in Fig. 1.

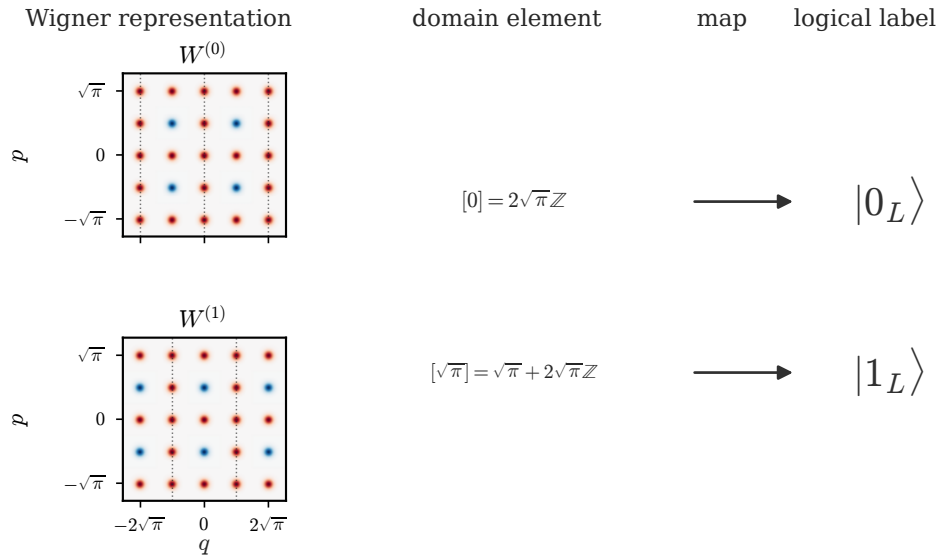


Figure 2. Quotient-lattice offset classes as logical GKP labels. The Wigner panels $W^{(0)}$ and $W^{(1)}$ show the two ideal codeword grids in Eq. (1.30); red and blue peaks denote opposite signs of the Wigner function. The dotted columns mark the q -coset whose position-eigenstate superposition forms the corresponding codeword. The domain elements $[0]$ and $[\sqrt{\pi}]$ map to $|0_L\rangle$ and $|1_L\rangle$, respectively.

Returning to the quotient description used to obtain Eq. (1.21), the logical label is assigned to the offset class of the position comb. Write

$$[q_0] = q_0 + 2\sqrt{\pi}\mathbb{Z}.$$

The two offset classes form the quotient

$$\sqrt{\pi}\mathbb{Z}/2\sqrt{\pi}\mathbb{Z} = \{[0], [\sqrt{\pi}]\} \cong \mathbb{Z}_2,$$

with the two classes given by

$$[0] = 2\sqrt{\pi}\mathbb{Z}, \quad [\sqrt{\pi}] = \sqrt{\pi} + 2\sqrt{\pi}\mathbb{Z}.$$

The label map is therefore

$$[0] \mapsto |0_L\rangle, \quad [\sqrt{\pi}] \mapsto |1_L\rangle.$$

In the Wigner representation, the same q -coset appears as a family of vertical columns. These columns mark the position eigenstates $|q = q_0 + 2s\sqrt{\pi}\rangle$ whose Dirac-comb superposition forms the logical codeword. Figure 2 places this quotient map next to the corresponding Wigner grids.

The Wigner function provides the visual language for the rest of this thesis. In the chapters that follow, we will see how physical imperfections deform this pristine lattice structure—broadening the delta peaks into finite-width Gaussians and contracting the grid toward the phase-space origin (Chapter 2). Recovery operations attempt to restore the lattice (Chapter 3), and the power-law relationship between energy and code quality (Chapter 4) manifests as sharper, more delta-like peaks at higher mean photon number.

But the ideal codewords $|0_L\rangle$ and $|1_L\rangle$ are non-normalizable states¹¹ with infinite mean photon number. Each is an infinite superposition of infinitely squeezed position eigenstates. Physically, the Dirac comb requires infinite energy because its peaks extend to arbitrarily large $|q|$. No physical oscillator can prepare them [1]. The physical implementations of GKP codes, including the construction of finite-energy approximations, and the dominant noise channel they face, is then the subject of Chapter 2.

¹¹ The norm $\langle 0_L | 0_L \rangle$ evaluates to an infinite sum of divergent $\delta(0)$ terms.

SECTION 2

Physical Realization of the GKP Code

The oscillator of Chapter 1 lives in an abstract infinite-dimensional Hilbert space. It's an ideal, purely algebraic construction. The next question is to implement it *physically*.

The GKP code is realized in the laboratory as the electromagnetic field of a superconducting microwave cavity. In present circuit quantum electrodynamic (cQED) experiments, this cavity is a high-quality-factor¹² aluminum resonator operated at dilution-refrigerator temperatures (~ 20 mK).

The specific device parameters relevant to this thesis are those of Sivak et al. [9], reporting a fundamental mode near $\omega_c/2\pi \approx 4.5$ GHz with a photon lifetime of $T_1^c = 606 \pm 10$ μ s. This physical setting already explains why the ideal GKP codewords of Chapter 1 cannot be realized. The cavity can store only a finite amount of energy, and the photons that store the logical state leak into the environment.

Constructing the abstract GKP code physically requires two coupled subsystems. A superconducting aluminum cavity stores the logical qubit; a transmon ancilla coupled to it supplies the anharmonicity needed for conditional control. Microwave drives at the cavity frequency ω_c implement logical gates directly on the oscillator mode; drives at the transmon frequency ω_q rotate and read out the ancilla. Their coupling is a dispersive cross-Kerr interaction that shifts each subsystem's frequency depending on the other's state without exchanging energy with the environment. Photon loss enters separately, through the cavity leaking to its electromagnetic surroundings at rate $\kappa = 1/T_1^c$, and is the central physical constraint of this chapter. The chapter develops each element in turn, moving from the physical platform and dispersive coupling (§2.1) through the finite-energy codewords (§2.2), logical gates and stabilizer measurements (§2.3), and the photon-loss channel (§2.4), to the energy–loss trade-off that determines the optimal squeezing parameter Δ (§2.5).

¹² The quality factor Q measures how many oscillation cycles the cavity completes before losing a significant fraction of its stored energy; a high- Q cavity retains its photons for many thousands of cycles, which is what makes it a viable quantum memory.

SUBSECTION 2.1

The Physical Setting

2.1.1 From harmonic cavity to anharmonic ancilla

Taken by itself, a single cavity mode is a harmonic oscillator (Eq. (1.2)) with equally spaced energy levels. Here “harmonic” means that every adjacent pair of Fock states $|n\rangle$ and $|n+1\rangle$ has the same transition frequency ω_c ; equivalently, the cavity is *linear* in the sense that driving one transition resonantly drives all of them. This property is useful for storing oscillator states (the cavity treats every Fock state democratically), but it is a problem for *control*.

Using the GKP code requires three capabilities: preparing finite-energy states, applying logical displacements, and measuring stabilizers to detect errors.

The first capability will return as a stabilization protocol after the control primitives are in place. The second is comparatively direct: displacements are driven by microwave pulses at the cavity frequency. But stabilizer measurement

requires a *conditional* operation: the cavity must respond differently depending on the state of an auxiliary system (§2.3). A harmonic cavity cannot do this, because all its transitions have the same frequency. We need an anharmonic system to serve as this auxiliary.

In superconducting circuits, the source of anharmonicity is the Josephson junction: a thin insulating barrier between two superconductors that acts as a nonlinear, dissipationless inductance [10].¹³ A superconducting circuit built from a Josephson junction shunted by a large capacitor is called a *transmon* [14]. Its Hamiltonian is

$$\hat{H} = 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\varphi}, \quad (2.1)$$

where $E_C = e^2/(2C_\Sigma)$ is the charging energy, E_J is the Josephson energy, $\hat{n}$ is the number of Cooper pairs transferred across the junction, and $\hat{\varphi}$ is the gauge-invariant superconducting phase difference [14]. The transmon operates in the regime $E_J/E_C \gg 1$ (typically $E_J/E_C \approx 50$), where the charge dispersion is exponentially suppressed and n_g can be set to zero [14].

¹³ “Josephson junction obeys quantum mechanics” was established by Devoret, Martinis, and Clarke in 1985, who demonstrated macroscopic quantum tunneling of the junction’s phase variable [11] and resolved the junction’s discrete, anharmonic energy levels [12]. These experiments were recognized with the 2025 Nobel Prize in Physics [13].

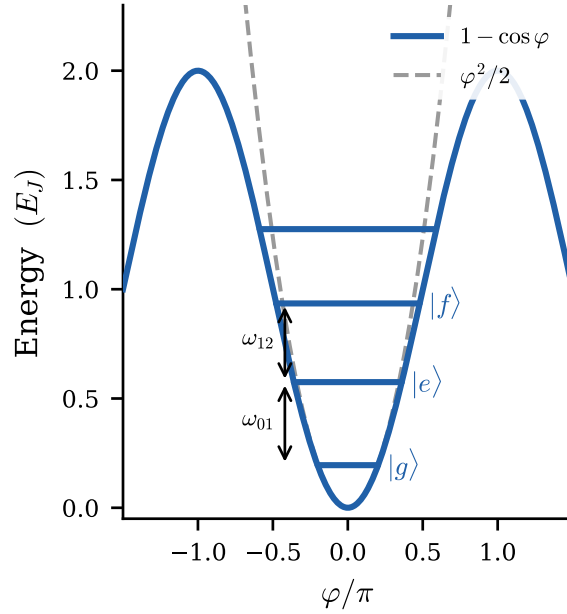


Figure 3. Energy levels of the transmon in the cosine potential $1 - \cos \varphi$ (solid) compared with the harmonic approximation $\varphi^2/2$ (dashed). The first four levels are computed perturbatively at $E_J/E_C = 50$ [14]. The decreasing spacing between adjacent levels, visible as $\omega_{12} < \omega_{01}$, is the anharmonicity that allows the lowest two levels to be addressed as a qubit without exciting higher transitions.

The cosine potential $-E_J \cos \varphi$ looks parabolic near its minimum at $\varphi = 0$, but flattens at larger phase excursions, and this deviation from a parabola is the origin of anharmonicity (Fig. 3). Expanding the cosine, $-\cos \varphi = -1 + \varphi^2/2 - \varphi^4/24 + \dots$, the φ^2 term yields a harmonic oscillator at the *plasma frequency* $\omega_p = \sqrt{8E_J E_C}/\hbar$, while the φ^4 term acts as a Duffing-type perturbation that progressively lowers each successive transition frequency.¹⁴ This quartic correction makes the transmon’s energy levels non-uniformly spaced, with an

¹⁴ The perturbative energy levels are $E_m \approx -E_J + \sqrt{8E_J E_C} (m + \frac{1}{2}) - \frac{E_C}{12} (6m^2 + 6m + 3)$ [14]. The quadratic dependence on m produces a uniform anharmonicity $\alpha = \omega_{12} - \omega_{01} = -E_C$.

anharmonicity $\alpha = E_{12} - E_{01} \simeq -E_C$, so each successive spacing is smaller than the last by an amount set by E_C . In typical devices, $|\alpha|/2\pi \approx 200\text{--}300\text{ MHz}$ [14], small compared to the transition frequency $\omega_q/2\pi \sim 4\text{--}6\text{ GHz}$, but large enough to single out the lowest two levels as a well-defined effective qubit. We label these $|g\rangle$ (ground) and $|e\rangle$ (excited).

In the context of GKP error correction, the transmon serves as an *ancilla*: an auxiliary quantum system used to control and measure the cavity mode, not to store the logical information itself. The logical qubit lives in the cavity; the transmon provides the nonlinearity needed to manipulate it.

2.1.2 Dispersive coupling and conditional control

The cavity and transmon interact because they share a capacitor, whose electric field couples to the charge on the transmon's capacitor plates and creates an interaction term in the joint Hamiltonian. When the transmon transition frequency ω_q is far from the cavity frequency (specifically, when the detuning $|\Delta| = |\omega_q - \omega_c|$ is much larger than the single-photon coupling strength g ¹⁵), the cavity and transmon do not resonantly exchange energy. Instead, the coupled system is described by an effective dispersive Hamiltonian, derived by Blais et al. [3]:

$$\hat{H}_{\text{disp}} \approx \hbar\omega'_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega'_q}{2} \hat{\sigma}_z + \hbar\chi \hat{a}^\dagger \hat{a} \hat{\sigma}_z, \quad (2.2)$$

where $\hbar\omega'_r \hat{a}^\dagger \hat{a}$ is the cavity energy at a *dressed* frequency ω'_r , shifted from its bare value ω_c by the interaction with the transmon. $\frac{\hbar\omega'_q}{2} \hat{\sigma}_z$ is the transmon energy at its dressed frequency ω'_q , where $\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|$ distinguishes the two ancilla states. The third term, $\hbar\chi \hat{a}^\dagger \hat{a} \hat{\sigma}_z$, couples the cavity photon number $\hat{a}^\dagger \hat{a}$ to the ancilla state $\hat{\sigma}_z$.

¹⁵ The vacuum Rabi rate g quantifies how strongly a single cavity photon couples to the transmon transition. It is the single-photon Rabi frequency of the Jaynes-Cummings interaction [3].

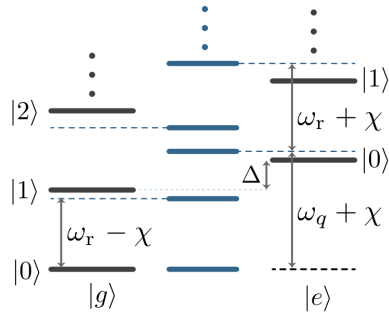


Figure 4. Energy spectrum of the dispersive Hamiltonian (2.2), showing dressed (blue) and uncoupled (gray dashed) levels. The two cavity ladders have state-dependent level spacings, $\omega'_r - \chi$ when the transmon is in $|g\rangle$ and $\omega'_r + \chi$ when in $|e\rangle$. Simultaneously, the transmon g - e transition frequency in the absence of cavity photons is the dressed value ω'_q . Adapted from Blais et al. [3], Fig. 9.

Writing $\hat{n} \equiv \hat{a}^\dagger \hat{a}$ for the photon number operator, the interaction term has two equivalent readings. Collecting cavity terms gives $\hbar(\omega'_r + \chi \hat{\sigma}_z) \hat{n}$, so the cavity frequency shifts by $\pm\chi$ depending on the transmon state ($+\chi$ for $|e\rangle$, $-\chi$ for $|g\rangle$). Collecting transmon terms gives $\frac{\hbar}{2}(\omega'_q + 2\chi \hat{n}) \hat{\sigma}_z$, so the transmon transition

frequency shifts by 2χ for each additional photon in the cavity. This photon-number-dependent frequency shift (sometimes called a cross-Kerr coupling) is the control resource that enables everything that follows. Once the cavity responds differently depending on the ancilla state, one can perform *conditional operations*, meaning operations on the cavity whose effect depends on whether the transmon is in $|g\rangle$ or $|e\rangle$.

Both readings are visible in the energy-level diagram (Fig. 4), where the two cavity ladders have different level spacings ($\omega'_r - \chi$ for $|g\rangle$, $\omega'_r + \chi$ for $|e\rangle$), while the transmon g - e transition frequency in the absence of cavity photons is the dressed value ω'_q .

Because the cavity frequency depends on the transmon state, a microwave drive pulse at the cavity frequency displaces the cavity by different amounts depending on whether the ancilla is in $|g\rangle$ or $|e\rangle$. This *conditional displacement*, in which the same drive pulse produces different phase-space translations conditioned on the ancilla state, is the single physical primitive that makes the GKP code operational. We develop its role in stabilizer measurement and state preparation in §2.3.

2.1.3 Hardware parameters

The device parameters relevant to this thesis are those of Sivak et al. [9], who reported the first GKP experiment to operate beyond *break-even*: the actively error-corrected logical qubit lasted longer than the best uncorrected encoding available in the same hardware.

The coherence gain is defined as $G = \Gamma_{\text{best passive}}/\Gamma_{\text{GKP}}$, where $\Gamma \equiv 1/T_{\text{logical}}$ is the inverse of the logical qubit lifetime (the timescale over which the encoded state retains fidelity); Γ_{GKP} is for the actively corrected GKP qubit and $\Gamma_{\text{best passive}}$ for the best uncorrected encoding on the same hardware. Here, the coherence gain G exceeded one.

Cavity lifetime	$T_1^c = 606 \pm 10 \mu\text{s}$
Cavity frequency	$\omega_c = 2\pi \times 4.479 \text{ GHz}$
Transmon lifetime	$T_1^t = 280 \pm 30 \mu\text{s}$
Dispersive shift	$\chi = 2\pi \times 46.5 \text{ kHz}$
QEC cycle time	$t_c = 9.85 \mu\text{s}$

At this cycle time the cavity loses a fraction $\gamma = 1 - e^{-t_c/T_1^c} \approx 0.016$ of its photons per error-correction round, about 1.6%. The physical hardware and its achieved quantum states are shown in Fig. 5: the aluminum cavity stores the oscillator mode, and the experimental Wigner functions (panel c) confirm that finite-energy GKP Pauli eigenstates at $\Delta = 0.34$ are preparable in this device.

Two consequences of the physical setting constrain what follows. First, the cavity stores only a finite amount of energy, so the ideal codewords must be replaced by finite-energy approximations with a squeezing parameter $\Delta > 0$. Second, photons escape at rate $\kappa = 1/T_1^c$, and each QEC cycle of duration t_c exposes the state to photon loss. We treat single-mode, square-lattice GKP under pure photon loss as the dominant error mechanism throughout the thesis.

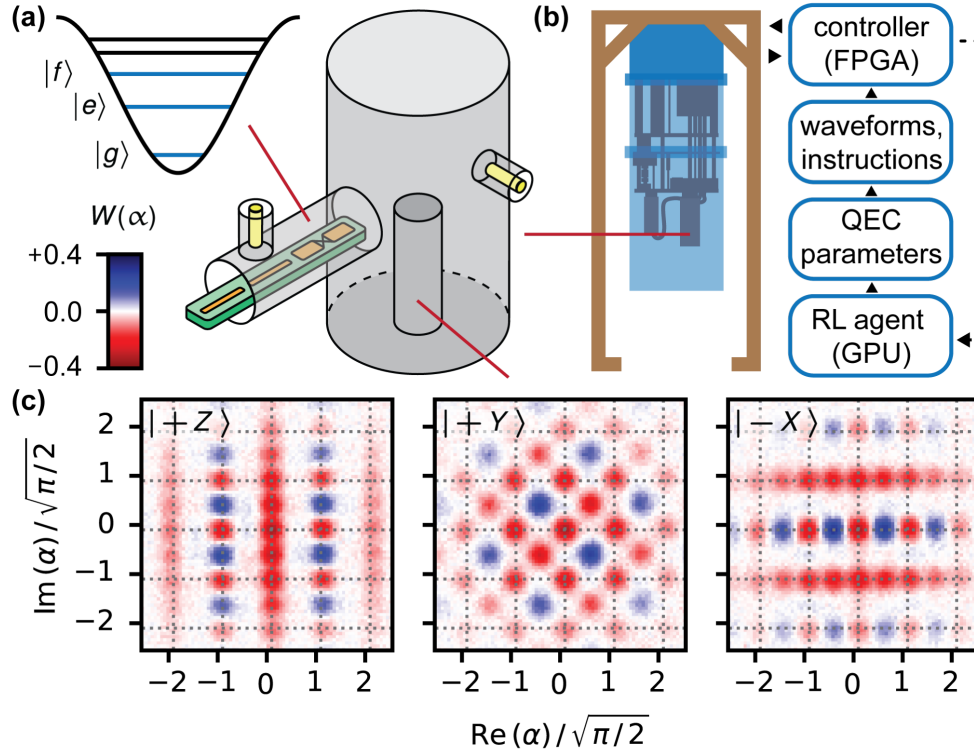


Figure 5. Experimental cQED system for GKP error correction. (a) A superconducting aluminum cavity stores the oscillator mode; a transmon circuit on a sapphire chip provides the anharmonic ancilla with energy levels $|g\rangle$, $|e\rangle$, $|f\rangle$ in the cosine potential (inset). (b) The sample is cooled in a dilution refrigerator and controlled with microwave and digital electronics; QEC parameters are optimized in situ by a reinforcement learning agent. (c) Experimental Wigner functions of the GKP Pauli eigenstates at $\Delta = 0.34$. Reproduced from Sivak et al. [9], Fig. 1.

SUBSECTION 2.2

Finite-Energy Codewords

The ideal codewords (1.21) are infinite superpositions of position eigenstates, each carrying a divergent norm. They cannot represent physical states in a laboratory cavity that stores finite energy. To make the GKP code physical, we replace each delta-function peak with a Gaussian of finite width Δ and impose a Gaussian envelope that suppresses distant peaks. The resulting states are normalizable, with a mean photon number that grows as the peaks sharpen.

2.2.1 Finite-energy construction

Gottesman, Kitaev, and Preskill [1] introduced finite-energy codewords by modifying the ideal Dirac comb in two ways.

First, each delta-function peak is replaced by a *squeezed vacuum* $|\psi_0\rangle$, a minimum-uncertainty Gaussian whose position variance $\text{Var}(\hat{q}) = \Delta^2/2$ is narrower than the vacuum value $1/2$, at the cost of increased momentum variance.¹⁶ The parameter Δ thus controls the width of each peak.

¹⁶ Squeezing a quadrature by a factor s means $\text{Var}(\hat{q}) = s^2/2$ and $\text{Var}(\hat{p}) = 1/(2s^2)$, preserving the minimum uncertainty product $\text{Var}(\hat{q})\text{Var}(\hat{p}) = 1/4$. The parameter Δ plays the role of s .

Second, a Gaussian envelope of width $1/\Delta_{\text{env}}$ weights the sum, suppressing peaks far from the phase-space origin. Each peak is placed at its lattice point by the position-translation operator $\hat{T}(u) = e^{-iu\hat{p}}$, which shifts a state by u along the position axis.¹⁷ For the logical-zero state, the construction gives

$$|\tilde{0}\rangle = \mathcal{N}_0 \sum_{s=-\infty}^{\infty} e^{-\Delta_{\text{env}}^2 (2s\sqrt{\pi})^2 / 2} \hat{T}(2s\sqrt{\pi}) |\psi_0\rangle, \quad (2.3)$$

where $\mathcal{N}_0$ normalizes the state. Each term in the sum places a squeezed Gaussian at position $2s\sqrt{\pi}$ (the even sublattice), weighted by the envelope factor $e^{-\Delta_{\text{env}}^2 (2s\sqrt{\pi})^2 / 2}$. Similarly,

$$|\tilde{1}\rangle = \mathcal{N}_1 \sum_{s=-\infty}^{\infty} e^{-\Delta_{\text{env}}^2 [(2s+1)\sqrt{\pi}]^2 / 2} \hat{T}((2s+1)\sqrt{\pi}) |\psi_0\rangle, \quad (2.4)$$

with peaks on the odd sublattice $(2s+1)\sqrt{\pi}$, just as the ideal codewords occupy even and odd multiples of $\sqrt{\pi}$ in (1.21). The construction has two independent parameters: Δ controls the width of each peak (the local Gaussian replacing a delta function), and Δ_{env} controls how quickly the envelope suppresses distant peaks (the global Gaussian weighting the sum). In the symmetric case $\Delta_{\text{env}} = \Delta$, the position- and momentum-space representations have the same form, with Gaussian peaks modulated by a Gaussian envelope. Since the Fourier transform inverts Gaussian widths, Δ governs peak sharpness in position space but envelope decay in momentum space, and vice versa for Δ_{env} [1]. We adopt this symmetric parametrization throughout.

In the cQED laboratory, these codewords are constructed via the small-big-small (SBS) stabilization protocol [9, 15]: sequences of ancilla-controlled conditional displacements and transmon resets iteratively steer the cavity toward the GKP subspace, sharpening the phase-space lattice over many rounds. The target squeezing Δ is set by the protocol parameters, and current experiments reach $\Delta \approx 0.34$ (9.4 dB) [9]. The mathematical structure of SBS is developed in §2.3.5.

The explicit construction (2.3)–(2.4) admits an equivalent single-operator form. In the symmetric case $\Delta_{\text{env}} = \Delta$, the combined effect of Gaussian peak broadening and Gaussian envelope suppression is reproduced by applying the photon-number damping operator $e^{-\Delta^2 \hat{n}}$ to the ideal codeword [5]. This equivalence, derived in Appendix C of [5], gives the compact definition used throughout the rest of the thesis:

Definition 4

(Finite-energy GKP codewords) For compact notation, write the two ideal codewords of Eq. (1.21) as $|\mu_L\rangle$ with $\mu \in \{0, 1\}$. Their finite-energy approximations are

$$|\mu_\Delta\rangle = \mathcal{N}_\mu(\Delta) e^{-\Delta^2 \hat{a}^\dagger \hat{a}} |\mu_L\rangle, \quad \mu \in \{0, 1\}, \quad (2.5)$$

where $\mathcal{N}_\mu(\Delta)$ is the normalization factor.

The envelope operator $\hat{E}_\Delta = e^{-\Delta^2 \hat{n}}$ is a photon number damping: it attenu-

¹⁷ This is the displacement operator (1.9) restricted to the real axis: $\hat{T}(u) = \hat{D}(u/\sqrt{2})$. It acts as $\hat{T}(u)|q\rangle = |q+u\rangle$.

ates each Fock component $|n\rangle$ by the factor $e^{-\Delta^2 n}$ [5, 15]. At $\Delta = 0$, the operator is the identity and we recover the ideal code. As $\Delta \rightarrow \infty$, every Fock component except $|0\rangle$ is suppressed and the state approaches vacuum. The family (2.5) therefore connects the formal ideal-code limit at $\Delta = 0$ to a trivial large- Δ limit in which both codewords are driven toward the oscillator vacuum.

2.2.2 Mean photon number

The mean photon number of the finite-energy codeword determines the energy cost of the encoding. Since $\hat{E}_\Delta = e^{-\Delta^2 \hat{n}}$ damps each Fock component $|n\rangle$ by $e^{-\Delta^2 n}$, the expectation

$$\langle \hat{n} \rangle = \frac{\sum_n n e^{-2\Delta^2 n}}{\sum_n e^{-2\Delta^2 n}}$$

reduces to differentiating a geometric series. The weights $e^{-2\Delta^2 n}$ form a geometric series with ratio $r = e^{-2\Delta^2}$, so $\sum_n e^{-2\Delta^2 n} = 1/(1-r)$ and $\sum_n n e^{-2\Delta^2 n} = r/(1-r)^2$, giving the leading term $\bar{n} = r/(1-r) = 1/(e^{2\Delta^2} - 1)$. The exact result [5] is

$$\bar{n} = \frac{1}{e^{2\Delta^2} - 1} + \delta, \quad (2.6)$$

where δ is exponentially small in $1/\Delta^2$, arising from the finite overlap between the two codewords. For small Δ the leading term dominates:

$$\bar{n} \approx \frac{1}{2\Delta^2} - \frac{1}{2}. \quad (2.7)$$

At the experimentally relevant value $\Delta = 0.34$, the exact formula gives $\bar{n} \approx 3.84$.¹⁸ Sharper peaks (smaller Δ) produce a better approximation to the ideal code but demand more energy, a trade-off we quantify in §2.5.

2.2.3 Intrinsic error and the Wigner function

For any $\Delta > 0$, the two codewords have a nonzero overlap $\langle 0_\Delta | 1_\Delta \rangle$, but it is exponentially small in $1/\Delta^2$ (of order 10^{-3} at $\Delta = 0.34$, from the suppression exponent $e^{-\pi(n_\Delta + 1/2)/d}$ [5]). A more consequential effect of finite squeezing is the intrinsic error it introduces in syndrome measurements, the classical outcomes of the stabilizer measurement that identify which decision cell the error falls in without revealing the logical state (§2.3 describes the full measurement circuit). When the stabilizer measurement bins a quadrature outcome to the nearest lattice point, the Gaussian tails of each peak can spill into the wrong decision cell (1.28). Each peak is a Gaussian of width Δ centered on a lattice point. The decision boundary lies at distance $\sqrt{\pi}/2$ from the center, so a misidentification occurs whenever the position displacement exceeds this boundary. The probability of this event is the Gaussian tail integral

$$P_{\text{tail}} = \int_{\sqrt{\pi}/2}^{\infty} \frac{1}{\Delta \sqrt{\pi}} e^{-q^2/\Delta^2} dq = \frac{1}{2} \operatorname{erfc}\left(\frac{\sqrt{\pi}}{2\Delta}\right),$$

The raw finite-energy codewords are not exactly orthogonal — their overlap is exponentially small ($\sim 10^{-3}$ at $\Delta = 0.34$) and is handled via a Löwdin-orthonormalized basis introduced in §3.1.

¹⁸ $\Delta = 0.34$ corresponds to squeezing $-10 \log_{10}(\Delta^2) \approx 9.4$ dB. The measured steady-state photon number under active QEC is $\langle \hat{n} \rangle = 4.67 \pm 0.02$ [9], higher than 3.84 because the state is not purely within the code space: the codespace occupation is $\langle \hat{\Pi}_0 \rangle = 0.82$.

where $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$ is the complementary error function, and the substitution $t = q/\Delta$ reduces the integral to standard form. For $\Delta \lesssim 0.5$ the argument $\sqrt{\pi}/(2\Delta)$ exceeds unity; applying the large-argument expansion $\operatorname{erfc}(x) \sim e^{-x^2}/(x\sqrt{\pi})$ and accounting for both cell boundaries gives [1]

$$P_{\text{err}} \sim \frac{2\Delta}{\pi} e^{-\pi/(4\Delta^2)}. \quad (2.8)$$

At $\Delta = 0.34$, this gives $P_{\text{err}} \approx 2.4 \times 10^{-4}$; at $\Delta = 0.25$, it drops below 10^{-6} [1]. The essential singularity $e^{-\pi/(4\Delta^2)}$ ensures that even modest improvements in squeezing yield exponential reductions in intrinsic error, but each reduction costs energy, since $\bar{n} \approx 1/(2\Delta^2)$ grows as the peaks sharpen.

These effects are directly visible in the Wigner representation. The envelope operator replaces each delta-function peak of the ideal Wigner function (1.30), the alternating-sign lattice of Chapter 1, with a Gaussian bump of width proportional to Δ , while the overall Gaussian envelope tapers the amplitude of distant peaks. The peak weight, which is $1/\pi$ per lattice site in the ideal case, drops to a finite value that decreases with increasing Δ . The checkerboard sign pattern persists, with the Wigner function retaining regions of negativity.

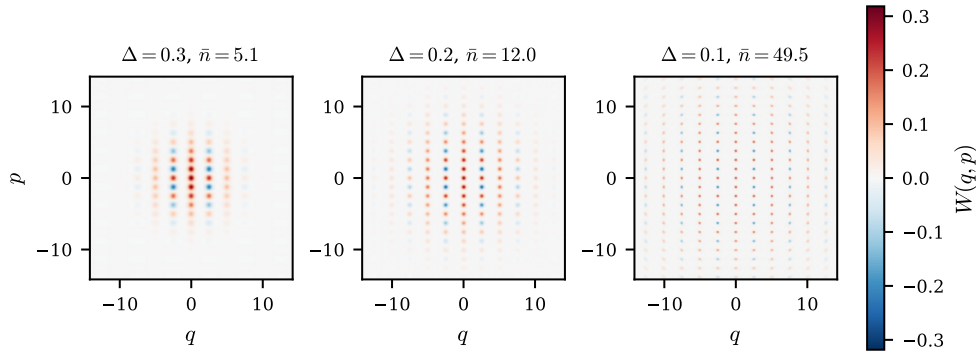


Figure 6. Wigner function of the finite-energy GKP codeword $|0_\Delta\rangle$ at three squeezing levels. As Δ decreases from 0.3 to 0.1, the peaks sharpen, the envelope broadens to include more lattice sites, and the state converges visually toward the ideal delta-peak lattice of Fig. 1. The grid spacing $2\sqrt{\pi}$ of the stabilizer period and the sublattice at $\sqrt{\pi}$ distinguishing $|0_\Delta\rangle$ from $|1_\Delta\rangle$ remain clearly visible at all three squeezing levels.

The convergence toward the ideal lattice is visible in the Wigner function (Fig. 6): the envelope operator provides a smooth, single-parameter interpolation between a physical state and the ideal code. This interpolation is the source of the energy–error trade-off that governs the rest of the chapter: every reduction in Δ sharpens the peaks and suppresses intrinsic error, but demands a higher mean photon number $\bar{n} \approx 1/(2\Delta^2)$, and higher-energy states are more vulnerable to photon loss.¹⁹

The basic physical implementation is now ready: a finite-energy state in a microwave cavity that approximates the ideal GKP code with controllable fidelity. The next question is how to operate this code by applying logical gates, detecting errors, and running the correction cycle.

¹⁹ Photon loss compresses the phase-space lattice toward the origin: a peak at $q_0 = 2n\sqrt{\pi}$ shifts to $\sqrt{\eta} q_0$ after transmissivity η , a displacement $(1-\sqrt{\eta}) \cdot 2n\sqrt{\pi}$ away from the nearest lattice point. This error grows with $|n|$, so states with support over many lattice sites are more vulnerable to loss-induced decoding errors. We quantify this in §2.5.

SUBSECTION 2.3

Operating the Code

The finite-energy codewords of the preceding section span a two-dimensional subspace of the oscillator Hilbert space. The remainder of this section develops the physical operations — logical gates, controlled displacements, stabilizer measurements, the QEC cycle, and the SBS preparation protocol — that make the code operational in hardware.

2.3.1 Logical gates as displacements

The logical Pauli operators of the GKP code are displacement operators (Eqs. (1.22)):

$$\bar{X} = e^{-i\sqrt{\pi}\hat{p}}, \quad \bar{Z} = e^{+i\sqrt{\pi}\hat{q}}.$$

The logical $\bar{Y} = i\bar{X}\bar{Z}$ follows from the Pauli algebra and requires no new primitive — it is implemented by sequencing the two displacements. Each shifts the cavity state by $\sqrt{\pi}$ along one quadrature axis. Physically, this is implemented by a single microwave drive pulse at the cavity frequency, with amplitude and duration chosen to produce the required phase-space displacement. No ancilla is needed. The cavity's linearity, which was a limitation for control in §2.1, is an advantage here: driving the cavity at its resonance frequency produces a clean displacement regardless of the photon number, so each logical Pauli gate requires only a single cavity-drive pulse.

2.3.2 Controlled displacements

Error detection, unlike logical gates, requires the cavity to respond differently depending on the state of the ancilla. A direct measurement of $\hat{q}$ would collapse the logical superposition, since the two codewords live at different positions; the syndrome must instead be extracted indirectly through ancilla-mediated operations, whose mechanism becomes clear once the controlled displacement primitive below is in hand (developed in §2.3.3).

The dispersive coupling (2.2) provides this bridge. Because the cavity resonance frequency shifts by $\pm\chi$ depending on whether the transmon is in $|g\rangle$ or $|e\rangle$, a microwave drive at the cavity frequency produces different displacements for the two ancilla states. The resulting operation, introduced by Campagne-Ibarcq et al. [2] as a *controlled displacement*, is

$$\hat{C}\hat{D}(\beta) = |g\rangle\langle g| \otimes \hat{D}(-\beta/2) + |e\rangle\langle e| \otimes \hat{D}(+\beta/2), \quad (2.9)$$

which shifts the cavity by $+\beta/2$ when the ancilla is excited and by $-\beta/2$ when it is in the ground state. The net differential displacement between the two branches is β .

Physically, $\hat{C}\hat{D}(\beta)$ is implemented by sending a microwave drive pulse to the cavity while the transmon is in a superposition. In the dispersive frame (2.2), the cavity frequency depends on the ancilla state: $\omega_c - \chi$ when the transmon is in $|g\rangle$, and $\omega_c + \chi$ when in $|e\rangle$. A drive at the bare cavity frequency ω_c is therefore detuned by $\mp\chi$ from the two branches. Over a drive time t , the accumulated

The displacement operator $\hat{D}(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$ was introduced in Eq. (1.9). It translates the oscillator state by α in phase space.

displacement in each branch is

$$|g\rangle : \hat{D}(-\chi t \beta_0), \quad |e\rangle : \hat{D}(+\chi t \beta_0),$$

where β_0 is set by the drive amplitude, so the differential displacement grows linearly with the dispersive shift χ [2]. In practice this naive implementation is slow, since the separation rate scales with $\chi/2\pi \approx 46.5$ kHz, and the echoed conditional displacement (ECD) gate of §2.3.5 replaces it in experiment, using a large intermediate coherent amplitude to boost the effective rate by over an order of magnitude [16]. This primitive, in which the cavity responds conditionally on the ancilla state, makes all GKP operations beyond simple displacements possible.

2.3.3 Stabilizer measurement

To detect displacement errors, we must measure the stabilizers S_q and S_p without destroying the encoded information. These stabilizers are periodic functions of the oscillator quadratures. Their eigenvalues repeat from one lattice cell to the next, so they can report the displacement within a cell without revealing the absolute lattice peak occupied by the state.

To see why, consider a codeword $|\psi\rangle$ that has been displaced by a small position error ϵ_q , so the state is $e^{-i\epsilon_q \hat{p}} |\psi\rangle$. The stabilizer $S_p = e^{i2\sqrt{\pi}\hat{q}}$ acts on this displaced state as

$$S_p e^{-i\epsilon_q \hat{p}} |\psi\rangle = e^{i2\sqrt{\pi}\epsilon_q} e^{-i\epsilon_q \hat{p}} |\psi\rangle, \quad (2.10)$$

where we used the BCH identity (1.12) in the form

$$e^{i\alpha\hat{q}} e^{-i\beta\hat{p}} = e^{i\alpha\beta} e^{-i\beta\hat{p}} e^{i\alpha\hat{q}}, \quad (2.11)$$

which follows from $[\hat{q}, \hat{p}] = i$, together with $S_p |\psi\rangle = |\psi\rangle$.²⁰ The displaced state is an eigenstate of S_p with eigenvalue $e^{i2\sqrt{\pi}\epsilon_q}$, a phase that encodes the position error. Crucially, this phase is unchanged when the displacement is shifted by any integer multiple of $\sqrt{\pi}$:

$$e^{i2\sqrt{\pi}(\epsilon_q + n\sqrt{\pi})} = e^{i2\sqrt{\pi}\epsilon_q} \cdot e^{i2\pi n} = e^{i2\sqrt{\pi}\epsilon_q} \quad \text{for any } n \in \mathbb{Z}.$$

Measuring S_p therefore reveals the position error only up to shifts by integer multiples of $\sqrt{\pi}$. This is the syndrome information needed to identify which decision cell (1.28) the error lies in, and no more. By the same argument, measuring $S_q = e^{-i2\sqrt{\pi}\hat{p}}$ reveals the momentum error ϵ_p only up to shifts by integer multiples of $\sqrt{\pi}$.

This periodic structure is why the measurement does not destroy the logical information. The two codewords $|0_L\rangle$ and $|1_L\rangle$ differ by a displacement of $\sqrt{\pi}$ in position, exactly one lattice spacing. Since the stabilizer eigenvalue is blind to shifts by integer multiples of $\sqrt{\pi}$, both codewords produce the same stabilizer outcome, so the measurement cannot distinguish them, and the encoded qubit is preserved [1, 7].

The measurement is implemented via a phase-estimation circuit [7]. The ancilla is prepared in a superposition $(|g\rangle + |e\rangle)/\sqrt{2}$, then a controlled dis-

²⁰ Explicitly:

$$\begin{aligned} S_p e^{-i\epsilon_q \hat{p}} &= e^{i2\sqrt{\pi}\hat{q}} e^{-i\epsilon_q \hat{p}} \\ &= e^{i2\sqrt{\pi}\epsilon_q} e^{-i\epsilon_q \hat{p}} e^{i2\sqrt{\pi}\hat{q}} \\ &= e^{i2\sqrt{\pi}\epsilon_q} e^{-i\epsilon_q \hat{p}} S_p. \end{aligned}$$

placement (2.9) entangles the cavity state with the ancilla. The relative phase accumulated between the $|g\rangle$ and $|e\rangle$ branches records the stabilizer eigenvalue, so reading out the ancilla extracts the syndrome [2]. If the measured error falls within the central decision cell, $|\epsilon_q| < \sqrt{\pi}/2$, it is unambiguously identified and can be corrected by applying the opposite displacement.

2.3.4 The error-correction cycle

One round of quantum error correction measures both stabilizer syndromes and applies the corrective displacement:

$$\text{measure } S_q \rightarrow \text{measure } S_p \rightarrow \text{correct.}$$

Each stabilizer measurement uses the ancilla-mediated protocol described above, consisting of a sequence of controlled displacements and ancilla rotations followed by a transmon readout whose binary outcome ($|g\rangle$ or $|e\rangle$) determines the corrective action [9]. A full QEC cycle, measuring both stabilizers, takes $t_c = 9.85 \mu\text{s}$.²¹

During each cycle, the cavity loses approximately 1.6% of its photons to the environment ($\gamma \approx 0.016$). The error-correction protocol must correct errors on a timescale short compared with photon-loss accumulation. This places a premium on the speed of conditional displacements, the bottleneck operation in each cycle.

2.3.5 Fast conditional displacements and state preparation

The bare dispersive interaction accumulates a conditional phase-space separation at a rate set by χ , which is slow compared to the decoherence time. The *echoed conditional displacement* (ECD) gate, introduced by Eickbusch et al. [16], overcomes this limitation by using a large intermediate coherent displacement as a lever arm.²² The gate implements

$$\text{ECD}(\beta) = \hat{D}(\beta/2) |e\rangle\langle g| + \hat{D}(-\beta/2) |g\rangle\langle e|, \quad (2.12)$$

Applied to a cavity state $|\psi\rangle$ with the ancilla prepared in $|+\rangle = (|g\rangle + |e\rangle)/\sqrt{2}$, the gate produces

$$\text{ECD}(\beta)(|\psi\rangle \otimes |+\rangle) = \frac{1}{\sqrt{2}} \left(\hat{D}\left(\frac{\beta}{2}\right) |\psi\rangle \otimes |e\rangle + \hat{D}\left(-\frac{\beta}{2}\right) |\psi\rangle \otimes |g\rangle \right),$$

entangling the two displaced copies of the cavity state with the ancilla. A subsequent ancilla measurement projects the cavity onto one branch, collapsing it to $\hat{D}(\pm\beta/2)|\psi\rangle$ depending on the outcome. Rather than waiting for χ to slowly build up this conditional separation, the protocol first displaces the cavity to a large coherent amplitude α_0 , where the same χ produces much faster state-dependent motion. The effective interaction rate becomes

$$g_{\text{eff}} = \chi \alpha_0, \quad (2.13)$$

²¹ The cycle decomposes as $t_c = 2 \times 4.924 \mu\text{s}$, with each half-cycle measuring one stabilizer [9].

²² A gate in this context is a unitary operation applied through a controlled, time-limited pulse sequence, an element of the unitary group acting on the joint cavity-transmon Hilbert space.

which can exceed the bare dispersive rate by more than an order of magnitude [16]. This speedup is what makes the QEC cycle fast enough relative to the photon-loss timescale.

Fast conditional displacements also enable state preparation. The small-big-small (SBS) protocol [9, 15] prepares and stabilizes finite-energy GKP states through a sequence of three conditional displacements per round. Each round implements the unitary [15]

$$\hat{U}^{(\text{SBS})} = e^{-i\epsilon\hat{p}\hat{\sigma}_y} e^{-ic_\Delta l\hat{q}\hat{\sigma}_x} e^{-i\epsilon\hat{p}\hat{\sigma}_y}, \quad (2.14)$$

where $l = 2\sqrt{\pi}$ is the lattice spacing, $\epsilon = s_\Delta l \approx \Delta^2 l$ is the small displacement scale, $c_\Delta = \cosh \Delta^2$, and $\hat{\sigma}_{x,y}$ act on the ancilla. The three factors are a second-order Trotter decomposition: the two “small” outer operations (conditional displacements of order Δ^2) sharpen the lattice peaks by projecting the cavity quadratures toward the nearest lattice point, while the “big” middle operation (a conditional displacement of order $\sqrt{\pi}$) trims the envelope by suppressing weight far from the origin [15]. In the circuit implementation, each exponential translates to a controlled displacement (2.9) sandwiched between ancilla rotations (Fig. 2 of [15]). After the three steps, the ancilla is reset to $|g\rangle$ by a fast measurement and conditional pulse. This reset is the dissipative element: it discards the ancilla’s entanglement with the cavity, projecting the cavity closer to the GKP subspace with each round. Over many rounds, the oscillator is iteratively steered from vacuum toward the target finite-energy GKP state with squeezing Δ , and the same protocol then maintains the state against drift. Current cQED experiments reach squeezing levels around 9–10 dB using this strategy [9].

With the operational picture in place, we turn to the dominant source of error.

SUBSECTION 2.4

Pure Photon Loss During the QEC Cycle

A real cQED system experiences several decoherence mechanisms: photon loss from cavity energy relaxation, cavity dephasing—often induced by transmon thermal excitations coupling through the dispersive interaction—transmon relaxation and dephasing, Purcell decay, and two-level-system defects [3]. For high-coherence 3D cavities, the energy relaxation rate κ associated with photon loss exceeds the pure dephasing rate κ_ϕ by roughly an order of magnitude [17], making photon loss the dominant error channel by a wide margin [3]. We therefore model each QEC cycle as a pure loss channel, deferring subdominant noise sources to the discussion in Chapter 5.

During each error-correction cycle of duration t_c , the cavity mode couples to the electromagnetic continuum outside its walls, and photons leak out at rate $\kappa = 1/T_1^c$. The net effect of this loss over one cycle is a quantum channel with transmissivity $\eta = e^{-t_c/T_1^c}$. We seek to model this irreversible process, but first, we have to develop the tools to describe a quantum system subject to noise and decoherence – an *open quantum system* – and the mathematical framework to represent the resulting quantum channels.

2.4.1 From closed to open evolution

In Chapter 1 the oscillator was an isolated system, fully described by a state vector $|\psi\rangle$ evolving under Schrödinger's equation. But a real cavity interacts with the electromagnetic environment outside its walls. When the cavity loses a photon to the environment, the cavity state becomes entangled with environmental degrees of freedom that we cannot track. The cavity alone is then no longer described by a pure state vector. It is instead a statistical mixture, and the appropriate description is the *density operator*

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|,$$

which encodes everything we can know about the system, with expectation values computed as $\langle\hat{O}\rangle = \text{Tr}(\hat{\rho}\hat{O})$. A pure state is the special case $\hat{\rho} = |\psi\rangle\langle\psi|$.

An irreversible process, one that cannot be undone by a unitary, maps density operators to density operators. Such a map is called a *quantum channel*, formally a completely positive, trace-preserving (CPTP) map $\mathcal{N} : \hat{\rho} \mapsto \mathcal{N}(\hat{\rho})$.²³ Trace preservation ($\text{Tr}[\mathcal{N}(\hat{\rho})] = 1$) ensures that probabilities are conserved. Every quantum channel admits an *operator-sum* (Kraus) *decomposition*:

$$\mathcal{N}(\hat{\rho}) = \sum_{\ell} \hat{E}_{\ell} \hat{\rho} \hat{E}_{\ell}^{\dagger}, \quad (2.15)$$

where the operators $\hat{E}_{\ell}$ satisfy $\sum_{\ell} \hat{E}_{\ell}^{\dagger} \hat{E}_{\ell} = \hat{I}$. The system and environment evolve jointly under a unitary $\hat{U}$, and tracing out the environment produces the operator sum (2.15). Each $\hat{E}_{\ell}$ corresponds to a particular outcome in the environment; for photon loss, $\hat{E}_{\ell}$ describes the event in which exactly ℓ photons are absorbed by the environment.

The beam-splitter model and its Kraus operators

The simplest physical model for photon loss treats the cavity mode $\hat{a}$ as mixing with a vacuum environment mode $\hat{b}$ on a beam splitter [4, 18]:

$$\hat{a} \rightarrow \sqrt{\eta} \hat{a} + \sqrt{1-\eta} \hat{b}, \quad (2.16)$$

where $\eta = e^{-\kappa t}$ is the *transmissivity*, the fraction of photons that survive after time t , and $\kappa = 1/T_1^c$ is the photon loss rate. The environment mode $\hat{b}$ starts in the vacuum state $|0\rangle_b$. After the beam-splitter interaction, the cavity and environment are entangled; tracing over the environment produces the Kraus operators derived in the next subsection. This picture makes the channel's two effects on phase space manifest. The system quadratures are scaled by $\sqrt{\eta}$ (contraction toward the origin), and the vacuum mode $\hat{b}$ adds noise (broadening). We quantify both effects in §2.4.2.

Tracing over the environment mode in the beam-splitter picture yields the Kraus operators of the pure loss channel [18, 19]. With the loss parameter $\gamma = 1 - \eta$, they are

²³ Complete positivity means that $\mathcal{N} \otimes \mathcal{I}$ remains a positive map for any identity extension $\mathcal{I}$, ensuring physical consistency even when the system is entangled with another system.

Definition 5 (Pure loss channel)

$$\hat{E}_\ell = \left(\frac{\gamma}{1-\gamma} \right)^{\ell/2} \frac{\hat{a}^\ell}{\sqrt{\ell!}} (1-\gamma)^{\hat{n}/2}, \quad \ell = 0, 1, 2, \dots \quad (2.17)$$

On Fock states, these act as

$$\hat{E}_\ell |n\rangle = \sqrt{\binom{n}{\ell}} (1-\gamma)^{\ell/2} \gamma^{(n-\ell)/2} |n-\ell\rangle, \quad (2.18)$$

removing exactly ℓ photons with a binomial probability distribution.

The full channel is

$$\mathcal{N}_\eta(\hat{\rho}) = \sum_{\ell=0}^{\infty} \hat{E}_\ell \hat{\rho} \hat{E}_\ell^\dagger. \quad (2.19)$$

The transmissivity $\eta \in [0, 1]$ parametrizes the channel strength. At $\eta = 1$ no photons are lost and the channel is the identity; at $\eta = 0$ every photon is absorbed and the output is vacuum. We also define the *loss depth* $x = -\ln \eta = \kappa t$, which parametrizes channel strength throughout the rest of the thesis.²⁴ For the Sivak experiment, $x = t_c/T_1^c \approx 0.016$ per QEC cycle.

²⁴ The loss depth is additive under concatenation: two successive loss channels with depths x_1 and x_2 compose to a single channel with depth $x_1 + x_2$.

Open Quantum dynamics and leading-order errors

The Kraus channel (2.19) describes the net effect of loss over a finite time interval t . The differential form, the continuous-time generator of this evolution, is the **Lindblad master equation**, the most general form of Markovian (memoryless) quantum dynamics [3]:

$$\frac{d\hat{\rho}}{dt} = \kappa \left(\hat{a} \hat{\rho} \hat{a}^\dagger - \frac{1}{2} \hat{a}^\dagger \hat{a} \hat{\rho} - \frac{1}{2} \hat{\rho} \hat{a}^\dagger \hat{a} \right). \quad (2.20)$$

The operator $\hat{a}$ appearing in the first term is called the *jump operator*; it annihilates one photon, representing a sudden quantum jump $|n\rangle \rightarrow |n-1\rangle$. The remaining two terms ensure that the density matrix stays properly normalized ($\text{Tr}[\hat{\rho}] = 1$ at all times). At the dilution-refrigerator temperatures of cQED experiments (~ 20 mK), the *thermal photon occupation*, the average number of thermal photons in the environment at the cavity frequency, is negligible ($\bar{n}_{\text{th}} < 0.01$), so the dominant process is photon loss into the vacuum. This small thermal occupation justifies the zero-temperature loss model [3].

The Kraus decomposition is exact but infinite. For the small loss per QEC cycle typical of current experiments ($\gamma \approx 0.016$), only the first two Kraus operators contribute significantly [18]:

$$\hat{E}_0 \approx \hat{I} - \frac{\gamma}{2} \hat{n}, \quad \hat{E}_1 \approx \sqrt{\gamma} \hat{a}. \quad (2.21)$$

The no-loss branch $\hat{E}_0$ applies a photon-number-dependent attenuation that slightly distorts the state even when no photon is lost. The single-loss branch $\hat{E}_1$

removes one photon, shifting the phase-space lattice in a manner that, projected onto the code structure, mimics a displacement error.

2.4.2 How loss distorts the GKP grid

The pure loss channel has a unique fixed point: the vacuum state $|0\rangle$. Every initial state is contracted toward vacuum under $\mathcal{N}_\eta$, with higher Fock components decaying faster. The beam-splitter picture (2.16) makes the channel's phase-space action transparent.

Recall that the beam splitter transforms the system mode as $\hat{a} \rightarrow \sqrt{\eta} \hat{a} + \sqrt{1-\eta} \hat{b}$, where $\hat{b}$ is a vacuum environment mode. In quadrature language this reads $\hat{q}_{\text{out}} = \sqrt{\eta} \hat{q} + \sqrt{1-\eta} \hat{q}_b$, and similarly for $\hat{p}$. Two consequences follow. First, the mean of each quadrature scales by $\sqrt{\eta}$: a Wigner function peak originally at (q_0, p_0) moves to $(\sqrt{\eta} q_0, \sqrt{\eta} p_0)$. The entire grid contracts toward the origin by the factor $\sqrt{\eta}$. Second, the environment mode adds vacuum noise.

Since $\hat{q}$ and $\hat{q}_b$ are independent,

$$\text{Var}(\hat{q}_{\text{out}}) = \eta \text{Var}(\hat{q}) + (1-\eta) \text{Var}(\hat{q}_b) = \eta \text{Var}(\hat{q}) + \frac{1-\eta}{2},$$

where we used $\text{Var}(\hat{q}_b) = 1/2$ for the vacuum mode. The added noise variance per quadrature is therefore $(1-\eta)/2$, corresponding to a Gaussian broadening of width $\sqrt{(1-\eta)/2}$ [4].

For the GKP grid, contraction and broadening erode the correctable region from opposite sides. The stabilizer peaks, originally separated by $2\sqrt{\pi} \approx 3.54$ in each quadrature, are pulled closer together, with the spacing reduced to $2\sqrt{\eta\pi}$ after loss. At the same time, each peak broadens by the added Gaussian width. The central decision cell (1.28), the square $[-\sqrt{\pi}/2, \sqrt{\pi}/2]^2$ within which errors can be unambiguously identified, is defined by the *original* lattice and does not shrink with the grid. A peak initially centered at q_0 is displaced by $(1-\sqrt{\eta})q_0$, so outer peaks move toward the wrong decision boundary faster than inner peaks.

To the level needed here, the pure loss channel contracts every Wigner function peak toward the origin by $\sqrt{\eta}$ and broadens each peak by convolution with a Gaussian whose width scales like $\sqrt{(1-\eta)/2}$ per quadrature. Contraction shrinks the grid; broadening smears its features. Both effects worsen as η decreases.

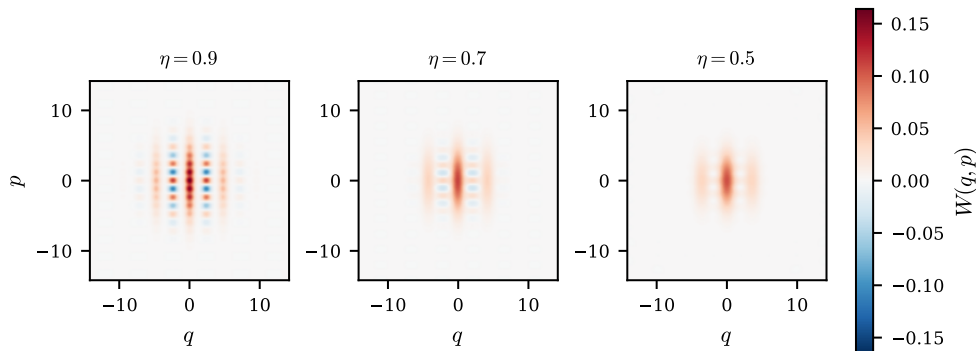


Figure 7. Wigner function of the finite-energy GKP codeword $|0_\Delta\rangle$ ($\Delta = 0.3$) after pure loss at three transmissivities. At $\eta = 0.9$, the grid structure is largely intact. At $\eta = 0.7$, the outer peaks have contracted toward the origin and the negative regions have begun to fill in. By $\eta = 0.5$, the peak spacing has shrunk by a factor of $\sqrt{0.5} \approx 0.71$ and the broadened peaks overlap significantly with the decision-cell boundaries.

The progression is visible in the Wigner function (Fig. 7). At $\eta = 0.9$, the grid structure is largely intact and the checkerboard negativity pattern is clearly resolved. At $\eta = 0.7$, the outer peaks have contracted noticeably toward the origin and the negative regions have begun to fill in. By $\eta = 0.5$, the peak spacing has shrunk by a factor of $\sqrt{0.5} \approx 0.71$ and the broadened peaks overlap significantly with the boundaries of the original decision cell, so syndrome measurements at this loss depth will frequently misidentify the error.

The loss channel does not destroy the grid structure instantaneously; it degrades it gradually. Higher-energy states therefore preserve sharper structure but incur a larger loss penalty.

SUBSECTION 2.5

Choosing the Operating Point

Sharper peaks (lower Δ , higher $\bar{n}$) produce a better code, since the intrinsic error (2.8) decreases exponentially with $1/\Delta^2$, and the Wigner function more closely approximates the ideal delta-peak lattice. But higher-energy states lose more photons per unit time, since the loss channel removes photons at a rate proportional to $\langle \hat{n} \rangle$. Why, then, do experiments stop around a finite squeezing level instead of pushing energy as high as possible?

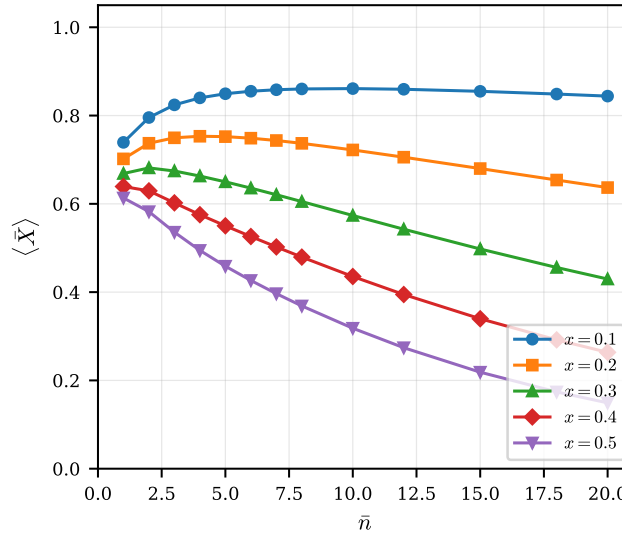


Figure 8. Baseline no-recovery behavior of the logical expectation value $\langle \bar{X} \rangle = \text{Tr}[\hat{D}(\sqrt{\pi}/2) \mathcal{N}_\eta(|+\Delta\rangle\langle+\Delta|)]$ as a function of mean photon number $\bar{n}$ at several loss depths $x = -\ln \eta$. At moderate loss, increasing $\bar{n}$ improves the logical observable; at higher loss the improvement saturates and eventually reverses as photon-loss damage overtakes the gain from sharper peaks. No recovery operation is applied in this figure.

The natural performance metric for a quantum channel acting on an encoded qubit is the *entanglement fidelity* F_E : one prepares half of a maximally entangled state, sends the other half through the channel, and measures the overlap with the original. $F_E = 1$ means the channel is perfectly reversible. Numerically, the entanglement fidelity of GKP codes under pure loss improves with $\bar{n}$ rather than

developing the finite-energy sweet spots seen in cat and binomial codes [18]. The ideal endpoint of this trend is controlled by a lower bound with an essential singularity at vanishing loss [18]:

$$F_E(\text{gkp}) > 1 - \exp\left(-\frac{\pi}{4c} \cdot \frac{1-\gamma}{\gamma}\right), \quad (2.22)$$

where c depends on the lattice geometry ($c = 2$ for the square lattice, $c = \sqrt{3}$ for the hexagonal lattice). As $\gamma \rightarrow 0$, the ideal infidelity vanishes faster than any power of γ , suppressed as $e^{-\text{const}/\gamma}$. The bound has no finite-energy parameter, so it establishes the asymptotic target rather than proving monotonicity in $\bar{n}$ by itself. The finite-energy corrections are developed by Zheng et al. [5] and used in Chapter 4. This bound assumes an optimal recovery operation; to visualize the raw effect of loss without recovery, Fig. 8 plots the unrecovered logical observable $\langle \bar{X} \rangle$ as a baseline diagnostic.

The energy-loss trade-off determines the optimal squeezing level for a given platform. Higher photons improve the code but accelerate loss proportionally. Sivak et al.'s operating point $\Delta = 0.34$ ($\bar{n} \approx 3.84$) achieves coherence gain $G = 2.27$, exceeding break-even [9] for their device ($T_1^c = 606 \mu\text{s}$, $\gamma \approx 0.016$). Pushing to higher $\bar{n}$ sharpens peaks but increases photon loss beyond what the QEC protocol can correct.

The trade-off is summarized in Fig. 8, which plots the logical expectation value $\langle \bar{X} \rangle$ as a function of $\bar{n}$ at several loss depths before any recovery is applied. At moderate loss ($x = 0.1$), increasing $\bar{n}$ produces steady improvement, since the exponential suppression of intrinsic error dominates the linear increase in photon loss. At higher loss ($x = 0.4$ – 0.5), the improvement flattens and eventually reverses, as the additional photons lost per cycle offset the gains from sharper peaks. This saturation marks the boundary of the regime where the unrecovered GKP state stops benefiting from additional energy.

The distorted grid can still be partially restored. The next chapter develops the recovery operations that attempt to reverse the effects of loss, and the theoretical benchmark against which we measure their success.

SECTION 3

Recovery of GKP States Under Loss

A GKP qubit that has passed through a lossy channel carries two kinds of damage. The grid has contracted toward the phase-space origin by a factor of $\sqrt{\eta}$, and each peak has broadened into the surrounding Voronoi cell (§2.4.2, Fig. 7). Without intervention, the logical information encoded in the grid degrades, and, as Fig. 8 quantifies, increasing the mean photon number eventually fails to compensate.

The stabilizer measurement introduced in §2.3.3 reveals the shift error modulo the lattice spacing $2\sqrt{\pi}$. But the loss channel does not simply shift the state. It rescales the entire grid toward the origin, a global transformation that moves every peak by a different amount. A peak originally at position q_0 moves to $\sqrt{\eta} q_0$, so the displacement needed to restore it depends on where the peak sits in phase space. No single corrective displacement can undo this damage across all peaks simultaneously.

Two complementary strategies address this problem. The first is operational: couple the damaged mode to a fresh ancilla, perform a joint measurement, and apply a correction conditioned on the outcome. This is a physical quantum error correction circuit that can run in a laboratory. The second is theoretical: construct a provably near-optimal recovery map that extracts close to the maximum fidelity from the damaged state, regardless of operational constraints. This recovery gives a benchmark against which any physical circuit can be measured.

Both strategies share a common structure: the qubit is encoded into the oscillator, subjected to loss, and then recovered. The composition of these stages — encoding, noise, recovery, decoding — defines a decoded logical map from the two-dimensional logical Hilbert space back to itself. Quantifying how well a recovery works reduces to measuring how close this map is to the identity, while keeping track of any weight that leaks out of the code space. We begin by establishing the formal framework that this map is built from.

SUBSECTION 3.1

The Logical Channel Framework

Before constructing or benchmarking a recovery, we need a precise language for encoding and decoding. The finite-energy codewords (2.5) span a two-dimensional subspace of the oscillator Hilbert space, and mapping logical states into and out of that subspace requires an encoding isometry, a code space projector, and a decoding map. Defining these objects is straightforward in principle, but one subtlety intervenes: the raw codewords $|0_\Delta\rangle$ and $|1_\Delta\rangle$ are not exactly orthogonal at finite Δ .

3.1.1 Orthonormalization

For any finite Δ , the Gaussian tails of peaks on the even sublattice overlap with those on the odd sublattice, producing a small but nonzero inner product

$\langle 0_\Delta | 1_\Delta \rangle$. Define the 2×2 Gram matrix

$$G_{\mu\nu} = \langle \mu_\Delta | \nu_\Delta \rangle, \quad \mu, \nu \in \{0, 1\}. \quad (3.1)$$

For finite Δ , G is positive definite with off-diagonal elements exponentially small in $1/\Delta^2$. At $\Delta = 0.34$, the overlap $|\langle 0_\Delta | 1_\Delta \rangle|$ is of order 10^{-3} [5] — negligible for Wigner plots and photon-number calculations, but nonzero and therefore incompatible with any definition that assumes an orthonormal logical basis.

The Löwdin symmetric orthonormalization [20] resolves this by constructing the orthonormal pair

$$|\phi_\mu\rangle = \sum_{\nu=0}^1 |v_\Delta\rangle (G^{-1/2})_{\nu\mu}, \quad \langle \phi_\mu | \phi_\nu \rangle = \delta_{\mu\nu}, \quad (3.2)$$

where $G^{-1/2}$ is the positive-definite inverse square root from the spectral decomposition of G .²⁵ Since $G \approx I_2$ at $\Delta = 0.34$, the orthonormalized states $|\phi_\mu\rangle$ are nearly indistinguishable from $|\mu_\Delta\rangle$ in practice; in simulations the Gram matrix correction is retained for numerical exactness.

3.1.2 Encoding and decoding maps

With an orthonormal basis in hand, the encoding and decoding maps can be written directly.

Definition 6

(Encoding and decoding maps) Let $\{|\phi_0\rangle, |\phi_1\rangle\}$ be the Löwdin-orthonormalized codewords (3.2). The **encoding isometry** is

$$V = \sum_{\mu=0}^1 |\phi_\mu\rangle \langle \mu|, \quad V^\dagger V = I_2, \quad (3.3)$$

mapping a logical state $\rho_{\text{in}} \in \mathbb{C}^{2 \times 2}$ to a physical state $\rho_{\text{enc}} = V \rho_{\text{in}} V^\dagger$ in the oscillator. The **code space projector** is

$$P_L = V V^\dagger = \sum_{\mu=0}^1 |\phi_\mu\rangle \langle \phi_\mu|, \quad (3.4)$$

the rank-2 projector onto the physical code space. The **decoding map** extracts the logical content of a physical state by projecting onto the code space and mapping back to $\mathbb{C}^{2 \times 2}$:

$$\Pi(\rho) = V^\dagger P_L \rho P_L V. \quad (3.5)$$

For any logical state ρ_{in} , encoding followed by decoding is the identity: $\Pi(V \rho_{\text{in}} V^\dagger) = \rho_{\text{in}}$.

²⁵ Löwdin orthonormalization is the unique procedure that produces an orthonormal set $\{|\phi_\mu\rangle\}$ minimizing the total squared distance $\sum_\mu \| |\phi_\mu\rangle - |\mu_\Delta\rangle \|^2$ from the original set [20]. It preserves the symmetry of the original states: if $|0_\Delta\rangle$ and $|1_\Delta\rangle$ are related by a half-lattice displacement, so are $|\phi_0\rangle$ and $|\phi_1\rangle$.

For ideal codewords ($\Delta = 0$), the encoding isometry reduces to $V = |0_L\rangle\langle 0| + |1_L\rangle\langle 1|$. At finite Δ , the Löwdin basis $\{|\phi_\mu\rangle\}$ ensures $V^\dagger V = I_2$ exactly, even though the raw codewords $|\mu_\Delta\rangle$ are not orthogonal.

These three objects — V , P_L , and Π — are the formal scaffolding that the rest of the chapter fills in. The Petz recovery map (§3.4) is defined in terms of the code space projector P_L and the noise Kraus operators. The decoded logical map (§3.6) assembles all four stages into a single composition $\tilde{\Lambda} = \Pi \circ \mathcal{R} \circ \mathcal{N}_\eta \circ E$,

where $E(\rho) = V\rho V^\dagger$ is the encoding step and $\mathcal{R}$ is the recovery. We turn now to the operational recovery strategy.

SUBSECTION 3.2

Knill-Type Error Correction

The operational strategy requires a circuit that takes a damaged GKP state and produces a corrected one. We work here with an idealized teleportation circuit, abstracting away the transmon-mediated implementation of §2.3. The damaged data mode is coupled to one half of an encoded GKP Bell pair, homodyne detection is performed on those two measured modes, and the unmeasured half of the Bell pair becomes the output mode. The measurement outcomes determine the corrective displacement applied to that output.²⁶ Because the logical information is transferred to a new GKP mode through measurement and feedforward, this is a *teleportation-based* protocol.

The elementary entangling operation inside this circuit is the SUM gate between the data mode (mode 1) and the measured Bell-pair mode (mode 2), introduced by Gottesman, Kitaev, and Preskill [1] and analyzed in detail by Conrad [6]. It is defined as

$$\hat{S}_{12} = e^{-i\hat{q}_1\hat{p}_2}, \quad (3.6)$$

and transforms the quadrature operators as

$$\hat{S}_{12}^\dagger \begin{pmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{p}_1 \\ \hat{p}_2 \end{pmatrix} \hat{S}_{12} = \begin{pmatrix} \hat{q}_1 \\ \hat{q}_2 + \hat{q}_1 \\ \hat{p}_1 - \hat{p}_2 \\ \hat{p}_2 \end{pmatrix}. \quad (3.7)$$

The position of mode 2 absorbs the position of mode 1, while the momenta rearrange so that mode 1 carries the difference $\hat{p}_1 - \hat{p}_2$. This is the continuous-variable analog of a CNOT gate, with mode 1 as control and mode 2 as target.²⁷

To see how the SUM gate separates the error from the data, consider a data mode that has been displaced by a position error ϵ_q . The input to the gate is $e^{-i\epsilon_q\hat{p}_1}|\psi\rangle_1$ together with the measured half of the GKP Bell pair. After the gate, the measured mode contains the combination $\hat{q}_2 + \hat{q}_1$, so measuring $\hat{q}_2$ yields an outcome $\tilde{q}$ that includes the data mode's position—and hence the error ϵ_q . The conjugate homodyne measurement supplies the corresponding momentum syndrome. The unmeasured Bell-pair mode carries the teleported logical state and waits for the corrective displacement.

The full Knill error correction circuit therefore prepares a fresh encoded GKP Bell pair, couples the damaged input to one Bell-pair mode, and performs homodyne detection on the two measured modes. The resulting pair of continuous outcomes $(\tilde{q}, \tilde{p})$ encodes the error syndrome modulo the lattice spacing $\sqrt{\pi}$. A corrective displacement $\hat{D}(\delta\alpha)$ is then applied to the remaining Bell-pair mode, with $\delta\alpha$ determined by rounding $(\tilde{q}, \tilde{p})$ to the nearest lattice point [7].

Remark

The Knill circuit is mathematically equivalent to a Steane error correction circuit followed by a corrective displacement [6]. The Steane circuit uses the same SUM gate and homodyne measurements, but does not apply a final correction. The

²⁶ Homodyne detection measures a single quadrature ($\hat{q}$ or $\hat{p}$) of the electromagnetic field by interfering the signal mode with a strong coherent reference (a local oscillator) at the same frequency. The outcome is a continuous real number, the instantaneous value of the measured quadrature, with precision limited by vacuum noise.

²⁷ The name “SUM gate” comes from the fact that it adds the control's position to the target's position, analogous to a CNOT adding the control bit to the target bit modulo 2.

Knill circuit adds a displacement by

$$\mathbf{s} = \frac{1}{\sqrt{2\pi}}(-\tilde{q}, \tilde{p})^T, \quad (3.8)$$

which shifts the output state to the correct lattice point identified by the syndrome. This equivalence anchors the teleportation-based Knill protocol to the more familiar Steane framework from the quantum error correction literature.²⁸

In circuit quantum electrodynamics, the SUM gate can be replaced by a beam-splitter interaction between two cavity modes, which is more natural for the dispersive coupling architecture of §2.1. The beam-splitter variant achieves lower logical error probability for realistic finite-energy ancillae [7, 21].

The circuit produces continuous homodyne outcomes $(\tilde{q}, \tilde{p})$. How do we map these to discrete corrections?

SUBSECTION 3.3

Syndrome Decoding

The homodyne measurement in the Knill circuit yields a continuous real number for each quadrature, but the corrective displacement must be a lattice translation. We need a map from $\mathbb{R}$ to the set of lattice corrections. This is the *decoding* step.

The simplest decoder is the *standard binning* (SB) decoder [1]. For each quadrature outcome $\tilde{q}$, we round to the nearest integer multiple of $\sqrt{\pi}$ and define the correction as the residual,

$$\delta q = \tilde{q} - \sqrt{\pi} \text{round}\left(\frac{\tilde{q}}{\sqrt{\pi}}\right). \quad (3.9)$$

This is the closest-point map on a one-dimensional lattice with spacing $\sqrt{\pi}$. The decoder assigns the error to whichever lattice cell the outcome falls in, and the correction displaces the state back to the cell center.²⁹

At high squeezing, the SB decoder is near-optimal for the noisy teleportation setting analyzed by Jafarzadeh et al. With damped finite-energy GKP auxiliary states at $\beta = 0.1$ (corresponding to approximately 10 dB of squeezing), the average gate fidelity of the SB decoder matches the optimized decoder to four decimal places [21]. This near-optimality justifies treating simple binning as the operational baseline, while the pure-loss benchmark below is handled separately by the Petz recovery map.

We now have an operational pipeline for GKP error correction. A Knill circuit extracts the error syndrome, and a binning decoder maps it to a correction. But how good is this pipeline? Is there a recovery operation that provably extracts the maximum information from the damaged state? We turn to this question next.

SUBSECTION 3.4

The Petz Recovery Map

The Knill circuit with standard binning is one recovery strategy. The space of all possible quantum recovery operations is vastly larger. We now ask whether

²⁸ The shift vector is denoted $\mathbf{s}$ rather than $\boldsymbol{\eta}$ to avoid collision with the transmissivity η used elsewhere in the thesis.

²⁹ For general GKP lattices, decoding reduces to the closest vector problem (CVP) on the dual lattice. For the square lattice in one mode, CVP is trivial—it amounts to rounding each coordinate independently. For multi-mode lattices, CVP is NP-hard in general [6], but efficient approximation algorithms exist for the structured lattices used in practice.

there exists a recovery map that is provably the best, or close to the best, for any code under any noise channel.

Before answering this question, we must establish what “recovery” means as a mathematical object. A noise channel $\mathcal{N}$ acts on the full oscillator Hilbert space. A recovery map $\mathcal{R}$ is a linear map that takes a (possibly damaged) state and returns a state. For $\mathcal{R}$ to represent a physically realizable operation, it must be *completely positive* and *trace-preserving* (CPTP). Complete positivity means that $\mathcal{R}$ maps valid quantum states to valid quantum states even when the system is entangled with an ancilla; trace preservation means that probabilities sum to one after the recovery. The composition $\mathcal{R} \circ \mathcal{N}$ defines a quantum channel from the code space to itself. The goal is to choose $\mathcal{R}$ so that $\mathcal{R} \circ \mathcal{N}$ is as close to the identity on the code space as possible.

The answer to our question is yes. The construction, introduced by Barnum and Knill [22] and reformulated by Ng and Mandayam [23] as the *transpose channel*, provides a universal near-optimal recovery map.³⁰

Definition 7

(Petz recovery map) Let $\{E_i\}$ be the Kraus operators of a noise channel $\mathcal{N}$, and let P_L (3.4) be the projector onto the code space. Define the image of the code space under the noise,

$$N_L = \mathcal{N}(P_L) = \sum_i E_i P_L E_i^\dagger.$$

On the support of N_L , the Petz recovery map has Kraus operators

$$R_i = P_L E_i^\dagger N_L^{-1/2}, \quad (3.10)$$

so that the recovered state is $\mathcal{R}(\rho) = \sum_i R_i \rho R_i^\dagger$ for states reachable from the noisy code space.

Each Kraus operator R_i acts in three steps. The inverse square root $N_L^{-1/2}$ reweights the damaged state so that each reachable direction in Hilbert space contributes proportionally to how strongly the noise populates it, ensuring the recovery does not over- or under-correct any error branch. The adjoint noise operator $E_i^\dagger$ then reverses the i -th error process, and the projector P_L maps the result back onto the code space. In this sense the Petz map gives a canonical reversal of the noise, constructed entirely from the noise operators and the code space structure.

For the pure loss channel of §2.4, the noise Kraus operators E_ℓ (2.17) are indexed by the number ℓ of lost photons, and the image $N_L = \sum_\ell E_\ell P_L E_\ell^\dagger$ encodes how loss maps the code space into the full Hilbert space. The Petz Kraus operators $R_\ell = P_L E_\ell^\dagger N_L^{-1/2}$ then provide a concrete, computable recovery for the GKP code under photon loss [5].

Remark

(Regularization) The inverse $N_L^{-1/2}$ is understood as a pseudo-inverse, since N_L may have zero eigenvalues corresponding to components of Hilbert space that are unreachable from the code space under loss. This support-restricted map is trace preserving on the reachable subspace; a completion on the orthogonal

³⁰ The name “Petz recovery map” comes from the Petz transpose map [24] in quantum information theory. In the approximate QEC literature, the terms “transpose channel,” “Petz map,” and “reversal operation” are used interchangeably.

complement can be added if one wants a CPTP map on the full oscillator Hilbert space. In numerical work, we replace $N_L^{-1/2}$ with $(N_L + \varepsilon I)^{-1/2}$ for a small regularization parameter $\varepsilon > 0$. The results in §3.6 use $\varepsilon = 10^{-10}$ and are stable across six orders of magnitude in ε .

The key property of the Petz map is its near-optimality. If the best possible recovery map achieves entanglement fidelity F^{opt} , then the Petz map achieves fidelity F_{Petz} satisfying [22, 25]

$$1 - F_{\text{Petz}} \leq 2(1 - F^{\text{opt}}). \quad (3.11)$$

The Petz map's infidelity is at most twice the optimal infidelity. For GKP codes under moderate loss, where the optimal infidelity is small, the gap between the Petz map and the true optimum is much smaller than this worst-case bound suggests.

The factor-of-two bound guarantees near-optimality, but it does not tell us how good the best recovery actually is. To quantify performance, we need a formula for the optimal fidelity itself.

SUBSECTION 3.5

Near-Optimal Fidelity

The object that encodes all information about how a noise channel couples to a code is the *QEC matrix*. For a code with logical basis states $\{|\mu_L\rangle\}$, where μ, ν label logical basis states, and noise Kraus operators $\{E_k\}$, where k, ℓ label Kraus operators, the QEC matrix has elements

$$M_{[\mu\ell], [\nu k]} = \langle \mu_L | E_\ell^\dagger E_k | \nu_L \rangle. \quad (3.12)$$

The indices $[\mu\ell]$ combine the logical label $\mu \in \{0, 1\}$ with the Kraus index ℓ , so M is a matrix of dimension $d_L N_K \times d_L N_K$, where $d_L = 2$ for a qubit code and N_K is the number of Kraus operators retained in the truncation.³¹

For a perfect code, the noise Kraus operators map distinct logical states to orthogonal error spaces, and the matrix M is block-diagonal with $M_{[\mu\ell], [\nu k]} = c_{\ell k} \delta_{\mu\nu}$. The QEC matrix of an approximate code records how far the actual code departs from this ideal. Its off-diagonal blocks measure the mixing between logical states induced by the noise, which is the source of logical errors that no recovery can fully undo.

The result, due to Zheng et al. [26], expresses the fidelity achieved by the Petz recovery map in terms of the QEC matrix.

³¹ For a perfect quantum error-correcting code, the QEC matrix is block-diagonal: $M_{[\mu\ell], [\nu k]} = c_{\ell k} \delta_{\mu\nu}$ for some coefficients $c_{\ell k}$. This is the content of the Knill–Laflamme conditions. The off-diagonal blocks of M measure how far an approximate code departs from this ideal, quantifying the mixing between logical states induced by the noise.

Theorem 2

(Near-optimal fidelity [26]) The entanglement fidelity achieved by the transpose channel (Petz recovery map) is

$$\tilde{F}^{\text{opt}} = \frac{1}{d_L^2} \|\text{Tr}_L \sqrt{M}\|_F^2, \quad (3.13)$$

where $(\text{Tr}_L B)_{\ell, k} = \sum_\mu B_{[\mu\ell], [\mu k]}$ denotes the partial trace over code-space indices, and $\|\cdot\|_F$ is the Frobenius norm.

Its practical advantage is computational. Computing the optimal recovery by semidefinite programming (SDP) requires solving an optimization over all CPTP maps, with cost scaling as $\tilde{O}((d_L N)^{5.246})$, where N is the physical Hilbert space dimension [26]. The near-optimal fidelity (3.13) requires only a matrix square root and a trace, with cost $O((d_L N_K)^3)$ in the number of retained Kraus operators N_K , and it comes with a two-sided guarantee [26],

$$\frac{1}{2}(1 - \tilde{F}^{\text{opt}}) \leq 1 - F^{\text{opt}} \leq 1 - \tilde{F}^{\text{opt}}. \quad (3.14)$$

The right inequality says that $\tilde{F}^{\text{opt}}$ is a lower bound on the true optimum. The left inequality says that the gap is at most a factor of two in infidelity, consistent with the Barnum–Knill bound (3.11). When the code performs well and the infidelity is small, the two bounds nearly coincide, making the Petz map effectively optimal.

For finite-energy GKP codes under pure loss, the QEC matrix has been computed analytically by Zheng et al. [5]. Each dual-lattice vector $\mathbf{x}$ contributes to the infidelity with a weight proportional to $e^{-\text{const} \cdot |\mathbf{x}|^2}$, so vectors farther from the origin are exponentially suppressed. The infidelity is therefore dominated by the shortest vectors of the dual lattice, and the leading-order scaling involves modified Bessel functions weighted by the loss parameter. The full expression is a multi-line sum over lattice vectors; we refer the reader to [5] for the details and state only the asymptotic result.

In the limit of infinite energy (perfect squeezing), the near-optimal infidelity converges to

$$\lim_{\hbar \rightarrow \infty} (1 - \tilde{F}^{\text{opt}}) = e^{-\pi(1-\gamma)/(2\gamma)}, \quad (3.15)$$

where γ is the loss rate [26]. This is an essential singularity at $\gamma = 0$: the infidelity vanishes faster than any power of γ as loss decreases. The exponent $\pi(1-\gamma)/(2\gamma)$ is four times larger than that of the Albert lower bound (2.22) established in Chapter 2, giving a correspondingly stronger asymptotic suppression of the infidelity.

With the fidelity formula in hand, we can now characterize the decoded logical map that results from encoding, loss, and Petz recovery.

SUBSECTION 3.6

The Decoded Logical Map

We now assemble the complete pipeline. Using the encoding isometry V (3.3) and decoding map Π (3.5) from §3.1, the four stages are:

1. **Encode:** $\rho_{\text{enc}} = V \rho_{\text{in}} V^\dagger$ embeds a logical state into the oscillator.
2. **Noise:** $\rho_{\text{noisy}} = \mathcal{N}_\eta(\rho_{\text{enc}})$ applies the pure loss channel (2.19).
3. **Recover:** $\rho_{\text{rec}} = \mathcal{R}(\rho_{\text{noisy}})$ applies the Petz recovery map (3.10).
4. **Decode:** $\rho_L = \Pi(\rho_{\text{rec}}) = V^\dagger P_L \rho_{\text{rec}} P_L V$ projects back to $\mathbb{C}^{2 \times 2}$.

The composition defines the *projected decoded map*,

$$\tilde{\Lambda} = \Pi \circ \mathcal{R} \circ \mathcal{N}_\eta \circ \mathcal{E}, \quad (3.16)$$

a trace-nonincreasing map from 2×2 logical density matrices to 2×2 logical operators, where $\mathcal{E}(\rho) = V\rho V^\dagger$.

The decoded state $\rho_L = \tilde{\Lambda}(\rho_{\text{in}})$ need not have unit trace: the projector P_L in the decoding step discards any component that has leaked out of the code space. The trace

$$w = \text{Tr}(\rho_L) \quad (3.17)$$

is the *codespace survival probability*, the fraction of the state that remains within the code space after noise and recovery. For a logical observable $O \in \{\bar{X}, \bar{Y}, \bar{Z}\}$, we distinguish the *leak-aware expectation*

$$\langle O \rangle_{\text{leak}} = \text{Tr}(O \rho_L), \quad (3.18)$$

which includes the leakage penalty, from the *conditional expectation*

$$\langle O \rangle_{\text{cond}} = \frac{\langle O \rangle_{\text{leak}}}{w}, \quad (3.19)$$

which post-selects on survival in the code space. The conditional expectation $\langle \bar{X} \rangle_{\text{cond}}$ is the diagnostic used throughout this section.³²

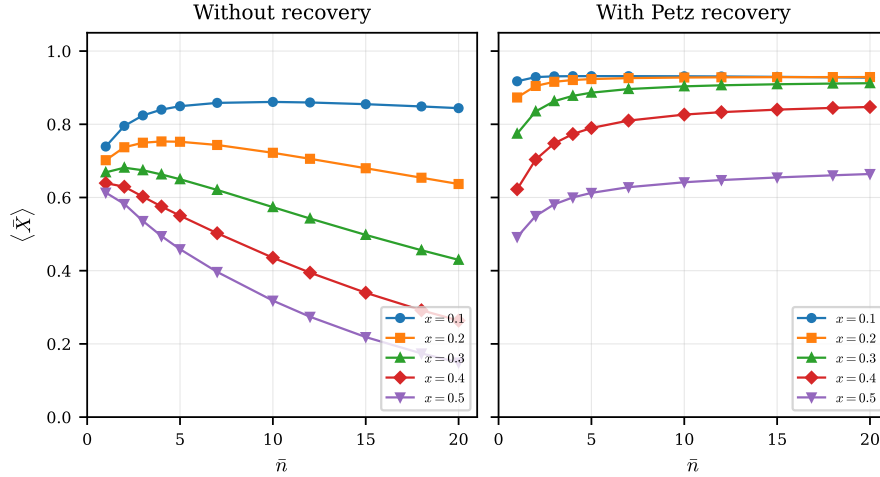


Figure 9. Conditional logical expectation value $\langle \bar{X} \rangle_{\text{cond}}$ for the input state $|+\Delta\rangle$ as a function of mean photon number $\bar{n}$ at five loss depths $x = -\ln \eta$. *Left:* without recovery ($\mathcal{R} = \text{id}$), reproducing the non-monotonic behavior of Fig. 8. *Right:* with Petz recovery $\mathcal{R}_{\text{Petz}}$, the logical observable remains high and approaches a loss-dependent plateau over the sampled energy range. At $x = 0.3$ and $\bar{n} = 20$, the recovered $\langle \bar{X} \rangle_{\text{cond}}$ reaches 0.91, compared to 0.43 without recovery.

The contrast between the unrecovered and recovered $\langle \bar{X} \rangle_{\text{cond}}$ is sharp. Without recovery (Fig. 8), $\langle \bar{X} \rangle_{\text{cond}}$ peaks at moderate $\bar{n}$ and declines at higher energy as photon loss overwhelms the sharpening of the peaks. With Petz recovery (Fig. 9), the high-energy collapse is removed over the sampled range and the

Under ideal GKP error correction with standard binning, the loss-induced logical channel is non-Pauli: the off-diagonal elements of its process matrix become appreciable at higher loss rates [27]. The same structure is expected for the Petz-recovered decoded map. This constrains the error-mitigation strategies of Chapter 4.

³² For a Pauli channel with equal X, Y, Z error rates, $\langle \bar{X} \rangle_{\text{cond}} = 2F_{\text{avg}} - 1$, where $F_{\text{avg}} = (d_L F_E + 1)/(d_L + 1)$ is the average fidelity and $d_L = 2$. The loss-induced map here is non-Pauli, so we report the conditional observable directly rather than converting it into a fidelity.

curves flatten toward a loss-dependent plateau.

Across the tested loss depths, recovery restores the broad intuition that added energy helps until the curves flatten. Increasing $\bar{n}$ sharpens the GKP peaks, reducing the intrinsic encoding error, and the Petz map reverses enough of the photon-loss damage that the unrecovered high-energy collapse is absent over the sampled range. The analytical bound (3.15) predicts that the infidelity approaches the essential singularity $e^{-\pi(1-\gamma)/(2\gamma)}$ from above as $\bar{n}$ grows, and the numerical curves are consistent with this loss-dependent plateau.

But the marginal gain diminishes. The curves in Fig. 9 flatten as $\bar{n}$ grows, and at high loss ($x = 0.5$), the recovered $\langle \bar{X} \rangle_{\text{cond}}$ saturates near 0.66 even at $\bar{n} = 20$. The asymptotic limit is set by the loss parameter alone, and no amount of additional energy can push the fidelity beyond it. If the logical observable at a sequence of energy levels follows a smooth curve, the trend itself contains information about the infinite-energy limit. The next chapter asks whether this trend can be exploited to estimate the code's asymptotic performance from finite-energy data alone.

SECTION 4

Energy-Scaled Zero-Noise Extrapolation

The Finite-Energy Gap

The ideal GKP code of Chapter 1 stores a logical qubit in an infinite-energy lattice; the finite-energy codewords of Chapter 2 broaden each peak by the envelope operator (2.5) and put the lattice within reach of a real device; the Petz recovery of Chapter 3 reverses enough of the photon-loss damage that the encoded qubit survives a lossy channel. These three constructions assemble into a single decoded-map pipeline, and the present chapter runs that pipeline end-to-end and turns it into a measurement instrument, extracting a quantity from finite-energy data that no individual run can reach.

The square-lattice GKP code with Petz recovery has an in-principle performance limit set entirely by the photon-loss strength of the channel. At infinite squeezing, the recovered logical fidelity approaches a value that depends only on the loss parameter $x = -\ln \eta$ and on no other hardware quantity. This in-principle limit is a property of the code-channel pair, computable from the code's lattice geometry and the channel's transmissivity, and it is what one would observe with arbitrarily high squeezing on an arbitrarily perfect device.

A real device with a finite squeezing budget produces a finite-energy codeword whose recovered fidelity sits below the in-principle limit by a gap that shrinks as the squeezing budget grows. Closing the gap by brute force requires more squeezing, the limiting resource on near-term cQED platforms.

Quantum error mitigation methods make estimates of the in-principle end of such a curve from data near the realized end, sampling an observable at several controlled noise levels and extrapolating to a noise-free target. They pay the cost in classical post-processing rather than in additional physical resources. Can we trade the energy limit of the finite-squeezing GKP code for that classical sampling overhead? To answer this question, we identify the noise that bounds the finite-energy GKP code, treat its mean photon number as the noise scale, sweep that scale across an energy ladder at fixed physical loss, and extrapolate the resulting sequence of recovered logical observables to the infinite-energy limit. The protocol holds wherever the code can correct the loss, extends naturally to two bosonic modes, and the squeezing savings it buys are largest where the code is most stressed.³³

SUBSECTION 4.1

Quantum Error Mitigation and Zero-Noise Extrapolation

Extrapolating a measured observable toward an unreachable noise level is a standard technique of quantum error mitigation (QEM). Its canonical instance is zero-noise extrapolation (ZNE) [28–31], which samples the observable at several values of a tunable noise knob and extrapolates the result to zero noise. Two ingredients are needed. The first is a knob whose effect on the expectation value is smooth; the second is a model for how the observable depends on the knob. In gate-based circuits, the standard knob is the evolution time, scaled

³³ Pairing a finite-distance bosonic code with classical post-processing of this kind sits within the early fault-tolerance programme, where software mitigation is stacked on imperfect hardware to approach algorithmic accuracy before full-scale quantum error correction is available. For a cavity with energy-relaxation time T_1 , a storage interval t corresponds to $x = t/T_1$; at the Sivak et al. parameters [9] ($T_1^c \approx 610 \mu\text{s}$, $t_c \approx 9.8 \mu\text{s}$ per QEC round), the depths $x \in \{0.2, 0.4\}$ we simulate correspond to roughly twelve and twenty-five consecutive QEC cycles.

by inserting identity-resolved gate sequences [29], and the standard model is a polynomial in the scale parameter recovered by Richardson extrapolation [28]. Neither applies directly to a bosonic code, because the noise that limits $\langle \bar{X} \rangle_{\text{cond}}$ is not of the gate-time kind.

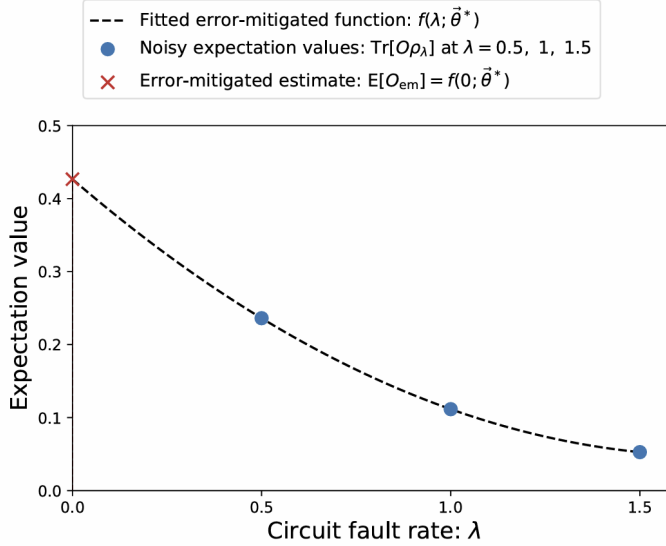


Figure 10. Schematic of standard zero-noise extrapolation, reproduced from Cai et al. [30]. An observable is measured at several controlled noise scales $\lambda \geq 1$, fit to a model in λ , and extrapolated to the noise-free point $\lambda = 0$. In gate-based implementations the noise scale is tuned by inserting identity-resolved gate sequences [29].

What limits the finite-energy GKP code is finite-squeezing error. The ideal code lives at $\bar{n} = \infty$, where the peaks of the Dirac comb are perfectly sharp and the stabilizers are exactly obeyed. At any finite $\bar{n}$, the peaks have been Gaussian-broadened by the envelope operator (2.5), the stabilizers are only approximately satisfied, and the resulting state carries a residual error that vanishes only as $\bar{n} \rightarrow \infty$. This residual is the noise we are extrapolating against, and $\bar{n}$ is its knob. The zero-noise direction of the axis is fixed by the physics rather than chosen by the analyst. The residual decreases as energy grows, so the zero-noise end of the axis is $\bar{n} \rightarrow \infty$, and a ladder of decreasing $\bar{n}$ plays the same amplification role as the gate-time scaling of the Richardson protocol.³⁴

SUBSECTION 4.2

The Extrapolation Target and Energy Schedule

Two results already in hand make the extrapolation well-posed. The asymptotic performance analysis of Albert et al. [18] establishes that the $\bar{n} \rightarrow \infty$ limit is finite and determined entirely by the loss parameter. The numerical evaluation of the near-optimal infidelity from the Bessel-series formula of Zheng et al. [5] shows that $\tilde{\epsilon}_{\text{opt}}(\bar{n}, x)$ decreases monotonically in $\bar{n}$ for every x below the channel capacity.

³⁴ A complementary QEM strategy, projective squeezing [32], virtually projects the output state onto a higher-squeezing subspace via a linear combination of unitaries. It requires only a single physical run with classical post-processing of measurement outcomes, whereas ES-ZNE requires repeated runs at multiple energies, and the two approaches are complementary rather than competing.

The target itself has a precise geometric origin. In the $\bar{n} \rightarrow \infty$ limit, the near-optimal infidelity of any GKP code is bounded above by a sum over the non-zero vectors of the symplectic dual lattice $\Lambda^\perp$ [5],

$$\lim_{\bar{n} \rightarrow \infty} \tilde{\epsilon} \leq \frac{1}{4} \sum_{\mathbf{x} \in \Lambda^\perp \setminus \{0\}} \exp \left[-\pi \frac{1-\gamma}{\gamma} |\mathbf{x}|^2 \right], \quad (4.1)$$

where $\gamma \equiv 1 - \eta$ is the loss fraction. Each dual-lattice vector contributes an exponentially decaying term, and the shortest vector dominates. For the square-lattice GKP qubit code ($d_L = 2$), the shortest non-zero vector satisfies $|\Lambda^\perp|_{\min}^2 = 1/d_L$ [5], and retaining only the leading contribution gives

$$\tilde{\epsilon}_\infty \approx \exp \left[-\frac{\pi}{d_L} \frac{1-\gamma}{\gamma} \right], \quad (4.2)$$

which for $d_L = 2$ reduces to the essential-singularity form (3.15) derived in Chapter 3. Below the channel capacity, $\tilde{\epsilon}_\infty$ is strictly positive and decreasing in η .

The approach to this target at finite energy is more structured than the single exponential of (4.2). Zheng et al. [5] expressed the near-optimal infidelity of the square-lattice GKP code as a Bessel series

$$\tilde{\epsilon}(\bar{n}, \gamma) = \frac{1}{d_L} \sum_{\mu, \nu} \sum_{n_1, n_2, m_1, m_2 \in \mathbb{Z}} e^{i\phi_{nm\mu\nu}} k^{|L_n|^2 + |L_m|^2} \mathcal{B}_{nm}(t, z), \quad (4.3)$$

in which the energy enters through $n_\Delta \equiv 1/(e^{2\Delta^2} - 1)$ and quantities derived from it, including the thermal factor $t = \gamma n_\Delta / (\gamma n_\Delta + 1)$, the exponential weight $k = e^{-\pi(n_\Delta + 1/2)}$, and the Bessel argument z . The lattice vectors L_n , phases $\phi_{nm\mu\nu}$, and Bessel functions $\mathcal{B}_{nm}$ carry the geometry of $\Lambda^\perp$, and the full closed form is stated in [5]. For the purposes of this chapter, Eq. (4.3) plays a structural rather than a fitting role. It shows that finite energy is organised by the single parameter $n_\Delta \approx \bar{n}$ and that the infinite-energy target is approached through a well-defined family of corrections, but it does not by itself determine the functional form of the recovered logical observable $\langle \bar{X} \rangle_{\text{cond}}$.

Turning $\bar{n}$ into a parameter we can dial requires mapping each target value $\bar{n}_j$ to the envelope parameter Δ_j that constructs the corresponding finite-energy codeword (2.5). The calibration condition is

$$\frac{1}{d_L} \text{Tr}(\hat{n} P_L(\Delta_j)) = \bar{n}_j, \quad (4.4)$$

where $P_L(\Delta_j)$ is the codespace projector (3.4) at squeezing level Δ_j and $d_L = 2$ for the single-mode qubit code. We solve (4.4) numerically by root-finding at each $\bar{n}_j$. Analytically, $\bar{n} \approx n_\Delta = 1/(e^{2\Delta^2} - 1) \approx 1/(2\Delta^2)$ for small Δ , so smaller Δ corresponds to sharper GKP peaks and larger mean photon number, as expected. The data points themselves are drawn from an arithmetic ladder,

$$\bar{n}_j = \bar{n}_0 - j \delta \bar{n}, \quad j = 0, 1, \dots, K-1, \quad (4.5)$$

with $\bar{n}_0 = 30$, $\bar{n}_{K-1} = 1$, and uniform spacing $\delta\bar{n} = 29/(K-1)$.

The resulting dataset at each loss depth is the raw material for the extrapolation. Figure 11 displays the conditional expectation $\langle \bar{X} \rangle_{\text{cond}}$ as a function of x for $\bar{n} = 1, \dots, 30$, with lighter shades marking low energy and darker shades marking high energy. Two qualitatively different regions share the plot. Below $x \approx 0.4$, the curves fan apart with $\bar{n}$, and every additional photon of energy delivers a measurable gain in logical coherence. Above $x \approx 0.4$, the ordering reverses and spending more energy actively hurts. The crossover near $x \approx 0.4$ marks the boundary between these two behaviours.

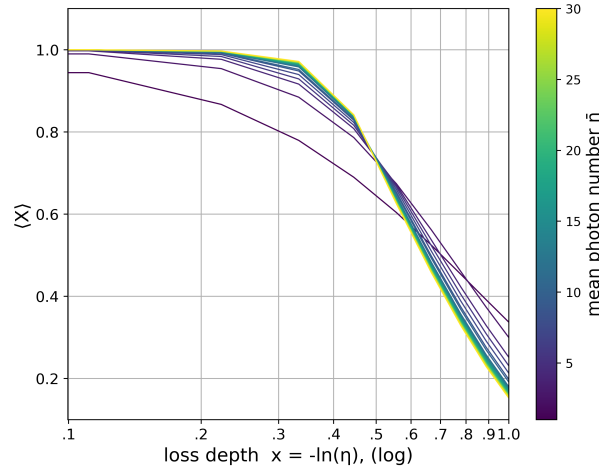


Figure 11. Conditional logical expectation $\langle \bar{X} \rangle_{\text{cond}}$ as a function of loss depth $x = -\ln \eta$ for thirty values of mean photon number $\bar{n} = 1, \dots, 30$ (lighter to darker). Below $x \approx 0.4$, $\langle \bar{X} \rangle_{\text{cond}}$ increases monotonically with $\bar{n}$; above it, higher energy degrades performance. The crossover near $x \approx 0.4$ marks the onset of the channel capacity limit and motivates the three-regime analysis of §4.4.

SUBSECTION 4.3

The ES-ZNE Protocol

At a single loss depth x , the ES-ZNE protocol runs the full four-stage decoded map (3.16) once per point on the energy ladder (4.5). At energy level j , we encode the logical input with the isometry $V(\Delta_j)$ (3.3), pass it through the pure-loss channel $\mathcal{N}_\eta$ (2.19), recover with the Petz map $\mathcal{R}$ (3.10), and decode via the projector Π (3.5).

For the single-qubit demonstration, we choose the logical input

$$\rho_{\text{in}} = |+_L\rangle\langle+_L|, \quad |+_L\rangle = \frac{|0_L\rangle + |1_L\rangle}{\sqrt{2}},$$

the $+1$ eigenstate of $\bar{X}$. This choice makes $\langle \bar{X} \rangle_{\text{cond}}$ a direct measure of how much logical X -basis coherence survives encoding, loss, recovery, and decoding. At finite energy, the encoded oscillator state is the corresponding $|+\Delta\rangle$.

The recorded quantity is the conditional expectation $\langle \bar{X} \rangle_{\text{cond}}(\bar{n}_j)$. Sweeping

across all K sampled energies yields the dataset

$$\{(\bar{n}_j, \langle \bar{X} \rangle_{\text{cond},j})\}_{j=0}^{K-1}.$$

Between sampled energies, only the envelope parameter Δ_j changes. The loss depth, the recovery map, and the decoding observable are all held fixed, so the “zero-noise” limit we extrapolate to is the infinite-energy limit $\bar{n} \rightarrow \infty$ at fixed physical loss, not vanishing physical loss.

The dataset is fit to

$$y(\bar{n}) = L + c \bar{n}^{-p}, \quad (4.6)$$

with three free parameters, the extrapolated infinite-energy limit L , the amplitude c , and the convergence exponent p . Only L is of physical interest; c and p are nuisance parameters of the fit. The observable being extrapolated is the conditional logical expectation (3.19) produced by the full decoded map (3.16), not the near-optimal infidelity of Eq. (4.3). We therefore do not treat the Bessel series as a derivation of (4.6). Instead, it justifies $\bar{n}$ as a structured finite-energy knob with a well-defined $\bar{n} \rightarrow \infty$ limit, while the actual warrant for the fit comes from the behaviour of the recovered observable itself.³⁵ In the protected and threshold regimes, the log-log residual is close to linear, so (4.6) is an effective low-parameter model for the observed convergence of $y(\bar{n})$ toward L . It trades analytical exactness for numerical robustness, sidestepping the non-trivial parameter estimation that fitting the full Bessel series would demand.

The adequacy of the model at a given x is checked by the log-log residual. If $y(\bar{n}) - L \approx c \bar{n}^{-p}$, then $\log |y(\bar{n}) - L|$ is linear in $\log \bar{n}$ with slope $-p$, and deviation from linearity is a signal that the power-law model has broken down. The residual, not the curve shape alone, warrants the fit. In the decoherence regime of §4.4, the residual ceases to be approximately linear, so the fitted power law no longer has a stable extrapolative meaning.

Uncertainty on the extrapolated parameter L is quantified non-parametrically by bootstrap resampling, refitting (4.6) on each draw and reporting the standard deviation of the resulting L distribution as the standard error. The procedure assumes nothing about the underlying noise model, but it captures sampling variance only. When the residual flags model breakdown, the systematic bias of fitting a power law to a non-algebraic curve is not reflected in the reported error, and the residual diagnostic and the confidence interval must be read together.

Figure 12 shows the full pipeline at $x = 0.2$. The left column traces the Wigner function of $|+\Delta\rangle$ as $\bar{n}$ grows from 1 to 30, each panel sharper than the one above it as the envelope tightens and the lattice peaks emerge. The extrapolation panel shows the corresponding fit to $\langle \bar{X} \rangle_{\text{cond}}(\bar{n})$, with the dashed line marking the inferred infinite-energy limit L . The state-evolution row fixes $\bar{n} = 4$ and follows the logical state through encoding, pure loss, and Petz recovery, so the visual sharpening of the lattice and the logical extrapolation appear as two views of the same pipeline.

The $|+\Delta\rangle$ panels have their own grid structure because the Wigner transform

³⁵ Alternative simple forms, including a pure exponential in $\bar{n}$, can mimic a monotone finite window, so the ansatz is not selected from appearance alone. The power law is kept only where the residual diagnostic below is linear for the recovered observable.

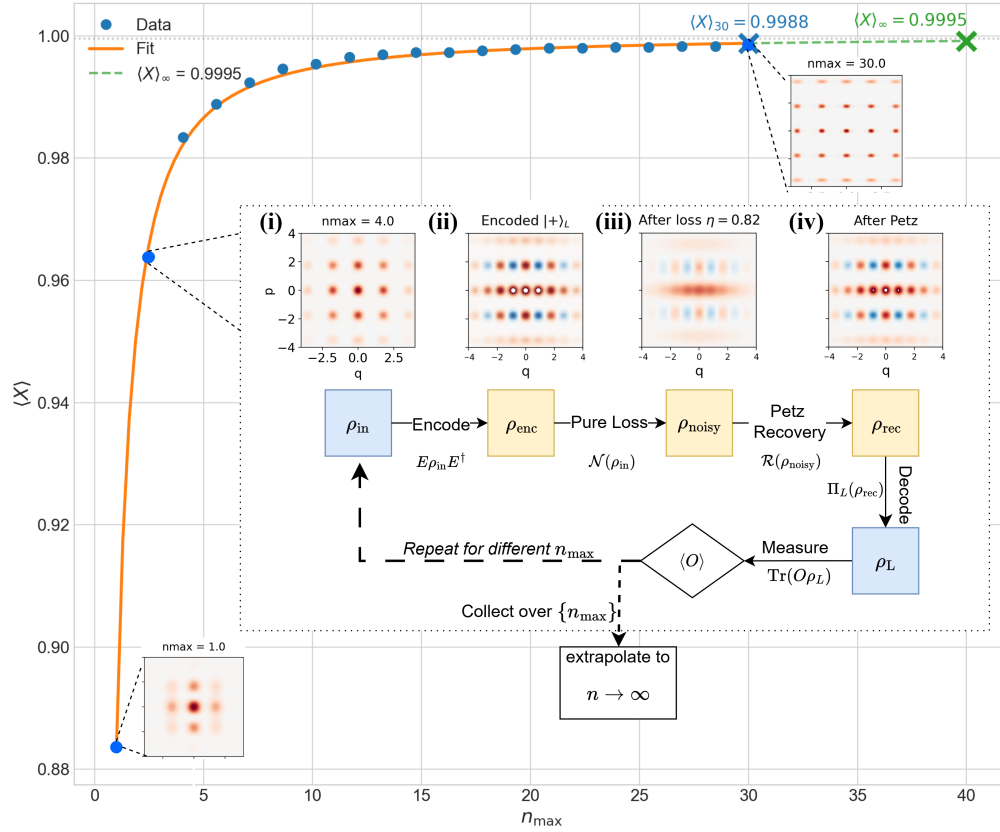


Figure 12. ES-ZNE pipeline at $x = 0.2$. Top left: Wigner snapshots of $|+\Delta\rangle$ at $\bar{n} = 1, 4, 30$. Top right: power-law fit to $\langle \bar{X} \rangle_{\text{cond}}(\bar{n})$ with extrapolated limit L . Bottom row: state evolution at $\bar{n} = 4$, from the maximally mixed codespace state and encoded logical $|+\rangle_L$ through pure loss and Petz recovery, together with the decoded-map workflow from ρ_{in} to the decoded state ρ_L .

acts on the density operator. In the ideal notation,

$$|+_L\rangle = \frac{|0_L\rangle + |1_L\rangle}{\sqrt{2}},$$

so

$$\rho_+ = |+_L\rangle\langle+_L| = \frac{1}{2}|0_L\rangle\langle 0_L| + \frac{1}{2}|1_L\rangle\langle 1_L| + \frac{1}{2}|0_L\rangle\langle 1_L| + \frac{1}{2}|1_L\rangle\langle 0_L|.$$

Since the Wigner transform is linear in the density operator,

$$W_+ = \frac{1}{2}W_0 + \frac{1}{2}W_1 + \frac{1}{2}W_{01} + \frac{1}{2}W_{10},$$

where W_{01} and W_{10} are the Wigner transforms of $|0_L\rangle\langle 1_L|$ and $|1_L\rangle\langle 0_L|$. The off-diagonal terms are therefore part of the plotted structure, and the finite-energy panels in Fig. 12 show the smoothed version of this same density-matrix decomposition.

SUBSECTION 4.4

Single-Qubit Results and Regime Analysis

Applying the ES-ZNE protocol across all simulated loss depths reveals a qualitative change in code behaviour as x crosses the crossover of Fig. 11. Three loss depths capture the arc of that change, and we walk through them before sweeping the whole range.

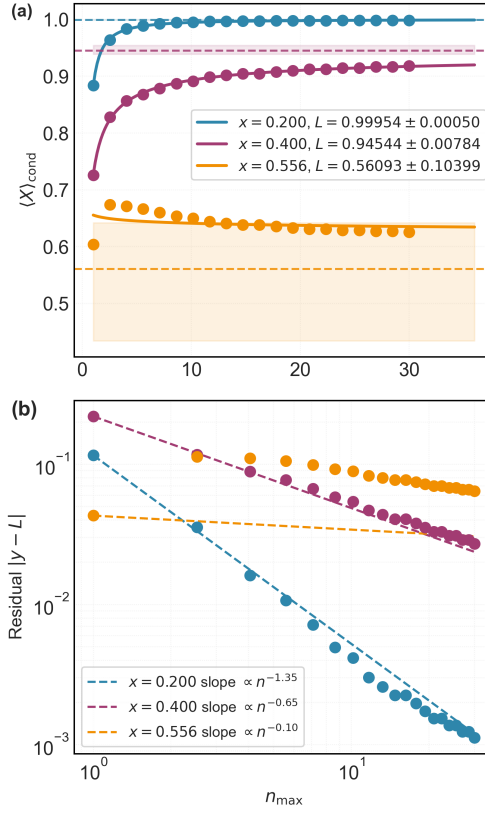


Figure 13. Power-law fits and log-log residual diagnostics for single-qubit ES-ZNE at three representative loss depths. (a): $\langle \tilde{X} \rangle_{\text{cond}}$ vs $\bar{n}$ (data points) with the power-law fit (4.6) (solid curve) and extrapolated limit L (dashed line). (b) $\log |\langle \tilde{X} \rangle_{\text{cond}} - L|$ vs $\log \bar{n}$; linearity indicates that the power-law model is adequate over the sampled window. At $x = 0.556$ the residuals are non-linear, indicating model breakdown in the decoherence regime.

At $x = 0.2$, deep in the low-loss regime, the power-law fit is excellent and the log-log residuals are linear. Over this sampled energy window, the model is adequate, and the extrapolated limit is $L = 0.99954 \pm 0.0005$, consistent with the ideal $\langle \tilde{X} \rangle = 1$ within numerical uncertainty. The best unextrapolated data point is 0.9988 at $\bar{n} = 30$, so ES-ZNE closes the remaining gap to one almost entirely.

At $x = 0.4$, near the capacity threshold, the power-law fit still tracks the data and the residuals remain approximately linear, so the same effective model is still adequate. The extrapolated limit, however, has fallen measurably below unity, marking the onset of a code limit set by the channel rather than by finite squeezing.

At $x = 0.556$, past the threshold, the trend reverses. Higher $\bar{n}$ yields *worse* performance. The Petz map, calibrated to the ideal codespace, displaces the loss-distorted state aggressively toward sharp lattice peaks that no longer match

the actual syndrome, which is a high-confidence error induced by excessive recovery [27]. The log-log residuals pull away from linearity.

Figure 13 places the three fits and their residuals side by side. The two low-loss panels lie close to a straight line in the residual view; the decoherence-regime panel curls away from it, visually marking the boundary of the window where the power-law model remains trustworthy.

Sweeping x from 0 to 0.8 and recording the extrapolated L at each depth produces the sigmoidal curve of Fig. 14, which organises the code’s operating region into three contiguous regimes.

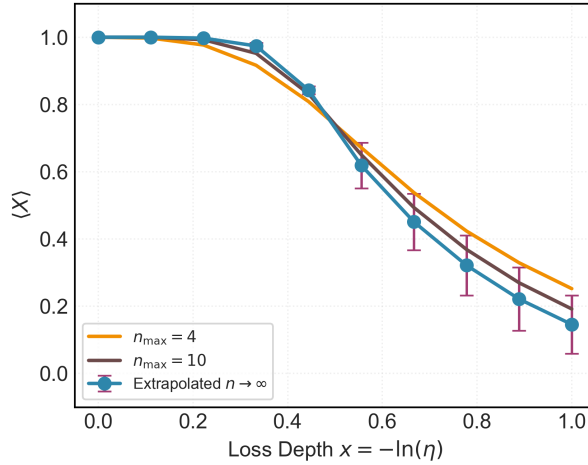


Figure 14. Extrapolated limit $L(x)$ as a function of loss depth (solid curve), compared with the unextrapolated $\langle \bar{X} \rangle_{\text{cond}}$ at $\bar{n} = 4$ (dashed) and $\bar{n} = 10$ (dotted). Shaded bands show bootstrap 1σ uncertainty. The three regimes are separated by the vertical dashed line at $x_{\text{crit}} \approx 0.4055$. In the protected regime, the extrapolated curve lies well above both baselines; in the decoherence regime, all curves converge.

In the *protected regime* ($x \lesssim 0.3$), the extrapolated limit sits at $L \approx 1$ with negligible uncertainty. The GKP code with Petz recovery achieves near-perfect logical coherence in the infinite-energy limit, and ES-ZNE removes the finite-squeezing artifact. In the *threshold regime* ($0.3 \lesssim x \lesssim 0.5$), L degrades continuously, tracking the channel-capacity limit. The quantum capacity of the pure-loss bosonic channel [5, 33] is

$$C_Q(\eta) = \max\left(0, \log_2 \frac{\eta}{1-\eta}\right), \quad (4.7)$$

and transmitting one logical qubit per bosonic mode at rate $R = 1$ requires $C_Q(\eta) \geq 1$. Setting $C_Q(\eta_{\text{crit}}) = 1$ gives $\eta_{\text{crit}} = 2/3$, or equivalently

$$x_{\text{crit}} = -\ln(2/3) \approx 0.4055. \quad (4.8)$$

This crossover is a property of the code-channel pair, and ES-ZNE estimates its location from finite-energy data.³⁶

In the *decoherence regime* ($x \gtrsim 0.5$), the extrapolated limit flattens at a low value with high uncertainty. The monotonic energy-coherence relation that underpins the entire protocol has broken down, so increasing $\bar{n}$ no longer im-

³⁶ The threshold $x_{\text{crit}} \approx 0.4055$ corresponds to roughly 25 QEC cycles at the Sivak et al. parameters [9]. The protected regime ($x \lesssim 0.3$, $\lesssim 19$ cycles) and the onset of the decoherence regime ($x \gtrsim 0.5$, $\gtrsim 31$ cycles) bracket a narrow device-relevant window.

proves performance at these depths and the extrapolation no longer gives a reliable estimate of the infinite-energy limit [30, 34].

Read together, the three regimes separate two distinct sources of logical error. The first is finite-squeezing error, the implementation-dependent gap between a finite-energy codeword and the ideal code. It vanishes as $\bar{n} \rightarrow \infty$, it is in principle reducible by investing more energy, and ES-ZNE estimates it away by extrapolation whenever the residual diagnostic remains linear. The second is the channel-capacity limit. Below η_{crit} , a single-mode GKP code with any recovery map cannot transmit one logical qubit per bosonic mode at unit fidelity, so crossing that threshold calls for a different code or architecture rather than more energy or more post-processing. In the decoherence regime, universal sampling lower bounds for QEM [35] and broader fundamental-limit analyses [30, 36] make the post-processing expensive in any case, consistent with the general impossibility of noise-free computation beyond the capacity threshold.

SUBSECTION 4.5

Two-Qubit Extension and Resource Analysis

The single-qubit analysis extends to two independent bosonic modes under the assumption of uncorrelated noise. The two-mode encoding is the product $V_{AB} = V_A \otimes V_B$, the noise channel factorizes as $\mathcal{N} = \mathcal{N}_{\eta_A} \otimes \mathcal{N}_{\eta_B}$, and the recovery factorizes similarly. The full two-mode decoded map is therefore

$$\Lambda_{AB} = \Lambda_A \otimes \Lambda_B,$$

where each Λ_i denotes the single-mode decoded map (3.16) of Chapter 3.

The product structure makes two-qubit expectation values computable without ever instantiating a two-mode joint simulation, provided each single-mode map is represented in a convenient basis. For factorized maps, the needed information is exactly how each single-mode decoded map acts on the Pauli operators. That basis is the Pauli transfer matrix.

Definition 8

(Pauli transfer matrix) For a single-mode decoded map Λ acting on $\mathbb{C}^{d_L}$, the Pauli transfer matrix is

$$\chi_{ij} = \frac{1}{2} \text{Tr}_L [\sigma_i \Lambda(\sigma_j)], \quad (4.9)$$

where $\sigma_0 = \mathbf{1}$, $\sigma_1 = X$, $\sigma_2 = Y$, $\sigma_3 = Z$ are the single-qubit Pauli matrices.

For a two-qubit input state $\rho_{\text{in}} = \frac{1}{4} \sum_{\mu, \nu} A_{\mu\nu} \sigma_\mu \otimes \sigma_\nu$ and a two-qubit observable $O_{ab} = \sigma_a \otimes \sigma_b$, the channel factorization gives

$$\langle O_{ab} \rangle_{\text{leak}} = \sum_{\mu, \nu} A_{\mu\nu} \chi_{a\mu}^{(A)} \chi_{b\nu}^{(B)}. \quad (4.10)$$

Only the four single-mode maps $\Lambda(\sigma_\mu)$, $\mu = 0, 1, 2, 3$, need to be computed per mode, and the full two-qubit expectation follows from the product $\chi^{(A)} \otimes \chi^{(B)}$. The factorization assumes separable recovery, with each mode's Petz map com-

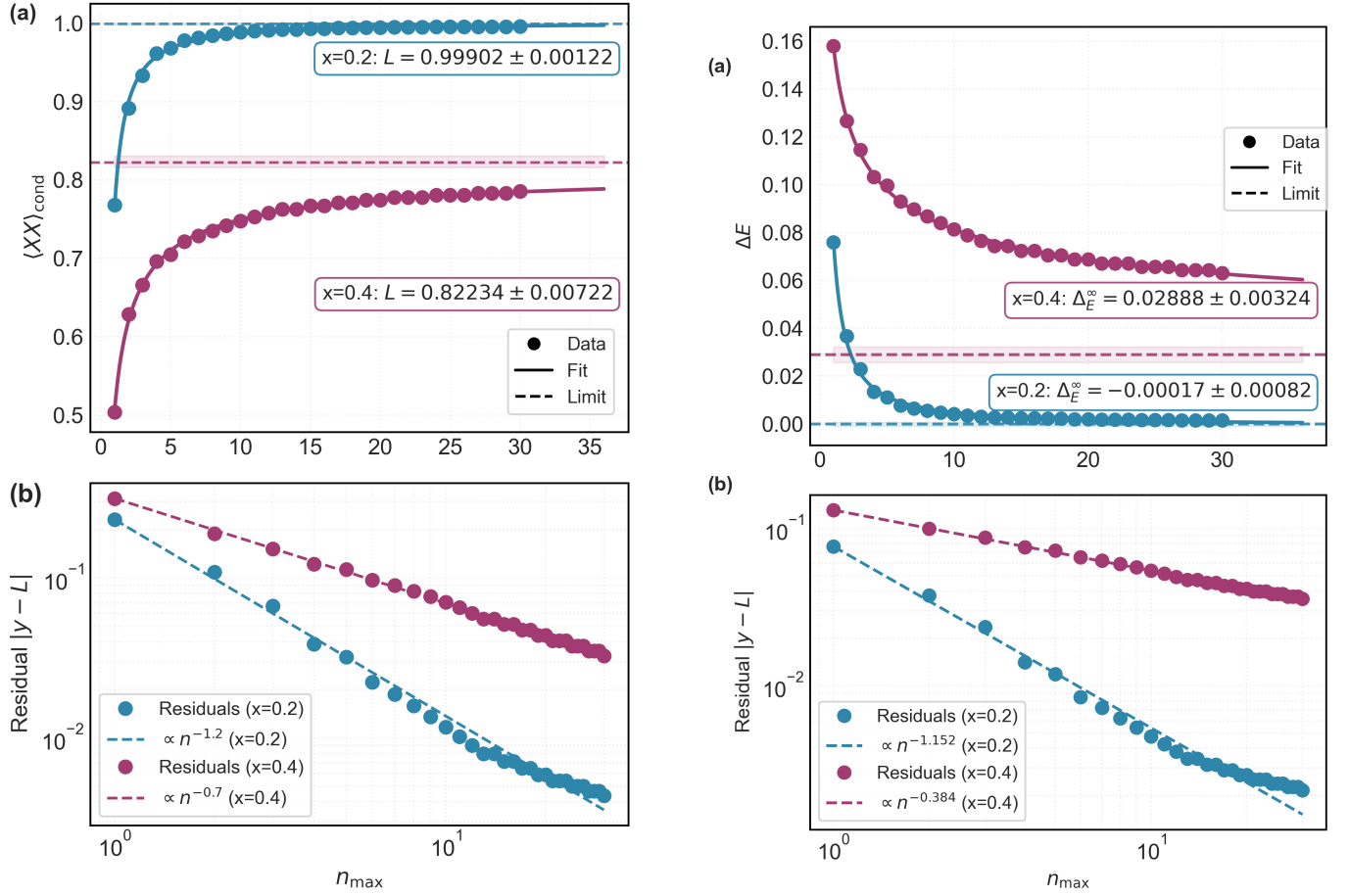


Figure 15. Two-qubit ES-ZNE. *Left:* Bell-state $\langle \bar{X}\bar{X} \rangle_{\text{cond}}$ vs $\bar{n}$ at $x = 0.2$ and $x = 0.4$ (top subpanel), with extrapolated limits $L = 0.99902 \pm 0.00122$ and $L = 0.82234 \pm 0.00722$; log-log residuals (bottom subpanel) are consistent with power-law behaviour over the sampled window. *Right:* Haar-random state-averaged mean absolute error $\Delta_E(\bar{n}, x)$ (Eq. (4.11)) over 50 random two-qubit pure states at $x = 0.2$ and $x = 0.4$; the power-law fit extrapolates to $\Delta_E^\infty \approx 0$ and $\Delta_E^\infty = 0.029 \pm 0.003$ respectively.

puted independently. Joint recovery strategies that exploit correlations across modes, relevant for entangled inputs or correlated noise, are left for future work. The cost of computing a two-qubit expectation value drops from exponential in the number of modes to linear, with four single-qubit pipeline runs per mode replacing an exponentially large joint simulation. ES-ZNE applies to any element of χ exactly as in the single-qubit case. We sweep $\bar{n}$ at fixed x , fit the ansatz (4.6) to each matrix element, and read off the extrapolated limit.

The first test of the two-mode pipeline is the maximally entangled logical state

$$|\Phi^+\rangle_L = \frac{|0_L\rangle|0_L\rangle + |1_L\rangle|1_L\rangle}{\sqrt{2}},$$

run under identical loss on both modes with the observable $\langle \bar{X}\bar{X} \rangle_{\text{cond}}$, which equals unity for the ideal state. Using the same energy ladder as in the single-qubit analysis, the extrapolation at $x = 0.2$ gives $L = 0.99902 \pm 0.00122$, consistent

with the ideal within uncertainty. At $x = 0.4$, it gives $L = 0.82234 \pm 0.00722$, reflecting the onset of partial decoherence across both modes simultaneously. In both cases the log-log residuals in the left panel of Fig. 15 remain linear, so the same power-law model remains adequate after the move to two modes.

The Bell state probes one direction in the two-qubit logical Hilbert space. A more comprehensive test averages over the whole space. We draw 50 random two-qubit pure states uniformly from the Haar measure on $\mathbb{C}^4$ [37], compute the conditional expectations of $\{XX, YY, ZZ\}$ at each state and energy level, and compare with the ideal zero-loss values. The state-averaged error is defined as

$$\Delta_E(\bar{n}, x) = \frac{1}{N_{\text{Haar}}} \sum_{r=1}^{N_{\text{Haar}}} \frac{1}{|S|} \sum_{O \in S} |\langle O \rangle_{\text{cond}}^{(r)}(\bar{n}, x) - \langle O \rangle_{\text{ideal}}^{(r)}|, \quad (4.11)$$

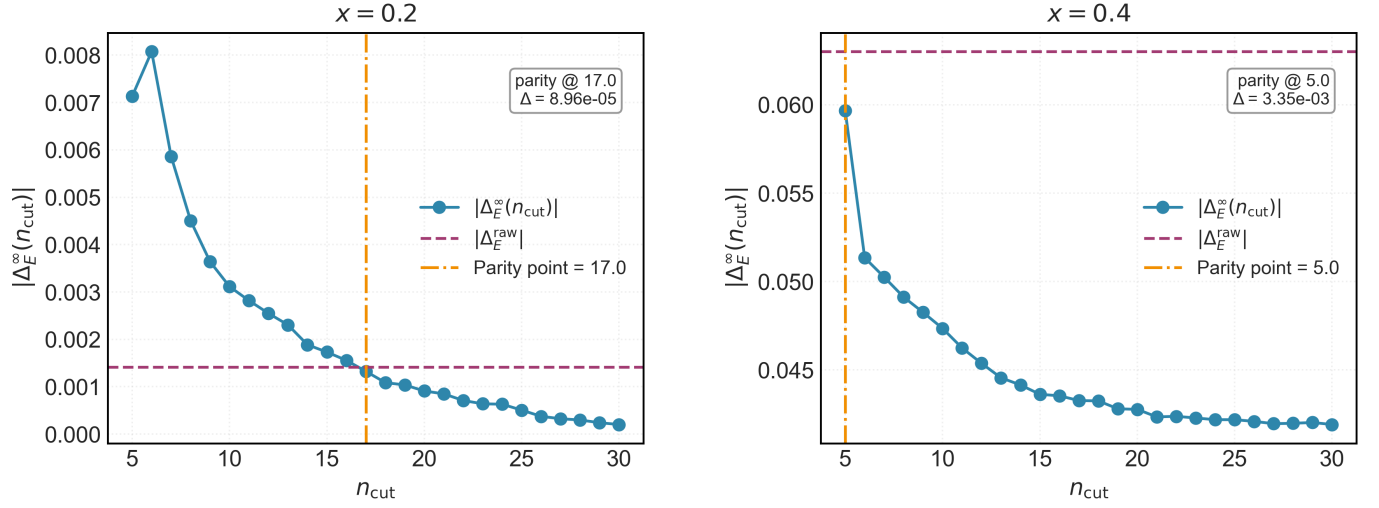
with $N_{\text{Haar}} = 50$ and $S = \{XX, YY, ZZ\}$. Applying ES-ZNE to Δ_E as a function of $\bar{n}$ produces an extrapolated state-averaged error Δ_E^∞ . At $x = 0.2$ we find $\Delta_E^\infty = -0.00017 \pm 0.00082 \approx 0$, near-ideal recovery across the full logical Hilbert space, with the negative central value reflecting finite fitting noise. At $x = 0.4$ we find $\Delta_E^\infty = 0.029 \pm 0.003$, a small but significant residual error consistent with the threshold onset already seen in the single-qubit sweep. The right panel of Fig. 15 shows the state-averaged extrapolation curves at both depths, and the same $\bar{n}$ -dependent fall-off that drove the single-qubit fits reappears for the full two-qubit error metric.

ES-ZNE reduces the squeezing demand.

We quantify it with a parity question: "What is the smallest energy cutoff $\bar{n}_{\text{cut}}$ such that an extrapolation using only data in $\bar{n} \leq \bar{n}_{\text{cut}}$ matches the *unextrapolated* accuracy at the full ladder maximum $\bar{n} = 30$?"

We vary $\bar{n}_{\text{cut}}$ in the power-law fit and track $|\Delta_E^\infty(\bar{n}_{\text{cut}})|$ against the raw benchmark $|\Delta_E^{\text{raw}}|$ evaluated at $\bar{n} = 30$. The parity point is the smallest cutoff at which the two quantities are equal.

At $x = 0.2$, parity is reached at $\bar{n}_{\text{cut}} \approx 17$. A sequence of low-energy simulations up to $\bar{n} = 17$ already matches the accuracy of an unextrapolated simulation at $\bar{n} = 30$, a 43% reduction in squeezing demand. At $x = 0.4$, parity is reached at $\bar{n}_{\text{cut}} \approx 5$, an 83% reduction. The savings are largest precisely where the code is most stressed, because finite-squeezing errors are largest near the threshold and the effective power-law tail is most useful there. The parity points are interpretable only while the log-log residual still certifies the power-law model. Figure 16 shows the parity analysis at both depths; in each panel the extrapolated curve crosses the raw benchmark well before the full ladder is reached.



(a) $x = 0.2$: parity at $\bar{n}_{\text{cut}} \approx 17$ (43% reduction).

(b) $x = 0.4$: parity at $\bar{n}_{\text{cut}} \approx 5$ (83% reduction).

Figure 16. Resource parity analysis. Absolute extrapolated error $|\Delta_E^\infty(\bar{n}_{\text{cut}})|$ (solid) as a function of the energy cutoff $\bar{n}_{\text{cut}}$ used in the power-law fit, compared with the raw benchmark $|\Delta_E^{\text{raw}}|$ at $\bar{n} = 30$ (dashed horizontal line). The vertical dash-dotted line marks the parity point where extrapolation matches unextrapolated accuracy at full energy.

Across the sampled ladder, the recovered observable is well described by a power law in $\bar{n}$ whenever the residual diagnostic stays linear, and in that window the extrapolation reduces the squeezing budget needed to reach a target accuracy. The parity analysis yields savings of 43% at $x = 0.2$ and 83% at $x = 0.4$. In the correctable regime, low-energy repeated runs can match a single higher-energy run.

The extrapolation method we developed in this chapter is conditional, not universal. It works here because pure photon loss gives a controlled channel, the square-lattice GKP qubit gives a recovered observable with a monotone finite-energy tail, and independent modes let the two-qubit extension factor through single-mode maps. Relaxing any one of these assumptions changes the object being extrapolated. Phase noise can deform the lattice without preserving the pure-loss Kraus structure, alternative GKP lattices change the dual-lattice geometry that sets the asymptotic limit, and other bosonic codes need not improve monotonically with energy. The usefulness of energy scaling therefore depends on which physical extensions preserve the same stable axis.

SECTION 5

Beyond the Thesis: Open Questions and Future Directions

Chapter 4 closed on the trade that ES-ZNE makes good. Wherever the GKP code of Chapters 2–3 is correctable, a low-energy ladder of repeated runs recovers the accuracy of a single high-energy run that no current hardware can supply.

The body of the thesis made three simplifications at three distinct moments. Chapter 1 chose the square lattice for consistency with the experimental platform and noted in passing that the same stabilizer construction defines a one-parameter family of qudit GKP codes at general d [1]. Chapter 2 modelled each QEC cycle as a pure loss channel, deferring subdominant noise sources to this chapter. Chapter 4 factorised the two-qubit ES-ZNE result through separable recovery on each mode, flagging correlated noise and joint recovery as future work.

Section 5.1 asks what happens when pure loss is no longer the only channel, and absorbs the correlated-noise question along the way. Section 5.2 asks what happens beyond the square-lattice qubit, where both alternative lattices and qudit codes live. Section 5.3 asks whether ES-ZNE ports to other bosonic code families, and finds that the monotone energy-fidelity curve GKP supplies is a specific structural property that other codes do not share.

SUBSECTION 5.1

Beyond Pure Photon Loss

The thesis models each QEC cycle as a pure loss channel because for high-coherence 3D cavities the intrinsic energy relaxation rate κ exceeds the pure dephasing rate κ_ϕ by roughly an order of magnitude, and photon loss dominates every other error channel by a wide margin [17].

The mixed loss-dephasing channel is not just the pure-loss channel with a small correction added afterward. Its loss and dephasing Lindbladians commute, so the channel can be written as loss followed by dephasing or dephasing followed by loss, but once both rates are nonzero the combined channel is not degradable [38]. The same analysis shows that optimized codes for the mixed channel need not saturate the energy constraint, and that the preferred known family crosses from hexagonal GKP at low dephasing to cat codes at high dephasing [38]. This is the first sign that leaving pure loss changes the code-design problem rather than merely adding another parameter to Chapter 2’s channel.

The structurally hardest subdominant channel for the GKP code is cavity phase noise. It enters through the Lindblad dissipator $\mathcal{D}[\sqrt{2\kappa_\phi}\hat{a}^\dagger\hat{a}]$, with jump operator $\hat{a}^\dagger\hat{a}$ non-linear in the quadratures, so the resulting channel is intrinsically non-Gaussian and breaks the closed-form Kraus structure on which Chapter 2 rests [17]. The deeper difficulty is that the square-lattice GKP code is less robust against phase noise than rotation-symmetric bosonic codes [17]. Stabilizer-subsystem simulations make the separation quantitative. At ten decibels of GKP squeezing, the square code can reach average gate infidelity below 10^{-3} for a loss rate near 4%, but only for a dephasing rate near 0.2% [39]. Decomposed into displacement operators, phase noise involves displacements of

arbitrarily large amplitude, so the lattice offers no fixed correctable window against it, and in the pure-dephasing limit cat and squeezed-cat codes outperform GKP for quantum communication rates [17]. Figure 17 shows what dephasing and the self-Kerr nonlinearity each do to a clean finite-energy GKP lattice.

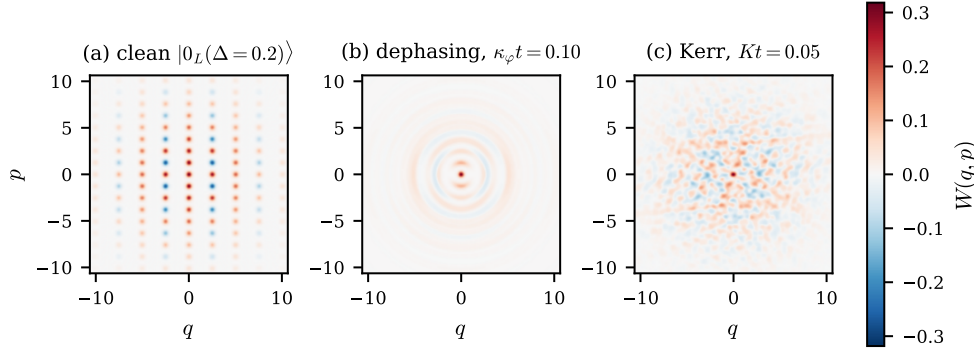


Figure 17. Subdominant noise channels deform a finite-energy GKP $|0_L(\Delta = 0.2)\rangle$ in qualitatively different ways than the pure photon loss of Section 2.4. *Left:* the clean reference. *Middle:* after pure-dephasing evolution at $\kappa_\phi t = 0.1$, the Fock off-diagonal coherences are damped by $\rho_{mn} \rightarrow \rho_{mn} e^{-\kappa_\phi(m-n)^2 t}$ and the lattice broadens. *Right:* after self-Kerr evolution $H_K = K \hat{n}(\hat{n} - 1)$ at $Kt = 0.05$, each Fock component acquires a phase $c_n \rightarrow c_n e^{-iKt n(n-1)}$, shearing the lattice features in phase space.

A second class of noise sources propagates from the dispersive ancilla into the oscillator, and these divide into coherent and incoherent errors [17]. The coherent errors are Hamiltonian distortions inherited from the nonlinearity used to prepare GKP states. The dispersive coupling leaves the oscillator with a parasitic self-Kerr of the form $H_K = K(\hat{a}^\dagger)^2 \hat{a}^2$, which rotates Wigner features in place and is generally harmful while the code is idling [17]. Multi-mode extensions inherit the same structure in cross-Kerr form $H_{ab} = \chi_{ab} \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b}$ between oscillators, and engineering a large on-off ratio between the intentional gate coupling and this parasitic interaction is itself an open problem [17]. The separable-recovery assumption of Section 4.5 therefore assumes that cross-Kerr is small enough to neglect during the storage interval, and once that assumption fails each mode’s Petz map must be replaced by a joint recovery.

The incoherent auxiliary errors flow through the same dispersive coupling in the other direction. Thermal excitation of the transmon imprints a random phase on the cavity through the dispersive shift $\chi \hat{a}^\dagger \hat{a} \hat{\sigma}_z$ and adds an effective oscillator dephasing rate [17]. Because the transmon’s intrinsic T_1 is typically shorter than the cavity’s, the dispersive coupling lets the oscillator inherit transmon decoherence and reduces the effective cavity T_1 [17]. Gate sequences contribute a third channel. During the ECD-based primitives of Section 2.3.5, bit flips and phase flips of the transmon propagate to the oscillator state, and Sivak et al. observe this directly in their real-time GKP experiment [9]. Isolated ancilla leakage events to $|f\rangle$ appear in the syndrome trace, most are reset within the same QEC cycle, and residual leakage to states above $|f\rangle$ persists at a measured rate of 1.3×10^{-4} per cycle. That rate is three orders of magnitude below the

physical error rate of $p_{\text{err}} = 0.13 \pm 0.02$ per cycle reported in the same experiment, so leakage is genuinely subdominant on current devices.

Even under the clean single-mode pure-loss model of Chapter 2, the loss-induced GKP logical channel is non-Pauli, with appreciable off-diagonal elements in its process matrix at higher loss rates [27]; the decoded map of Chapter 3.6 inherits this complication. A joint recovery across correlated modes therefore inherits non-Pauli structure on each mode in addition to whatever cross-Kerr correlation it is asked to undo.

Each channel removes a different input to the derivation: phase noise the pure-loss Kraus structure, thermal dephasing and reverse Purcell the iid-per-cycle factorisation, self-Kerr the pure-loss Petz map, and cross-Kerr the tensor-product channel decomposition.

SUBSECTION 5.2

Beyond the Square-Lattice Qubit Code

Chapter 1.3 derived the GKP code from the area condition on the stabilizer pair, which admits any symplectically integral lattice. The square lattice was chosen for simplicity and consistency with the experimental platform of Chapter 2.

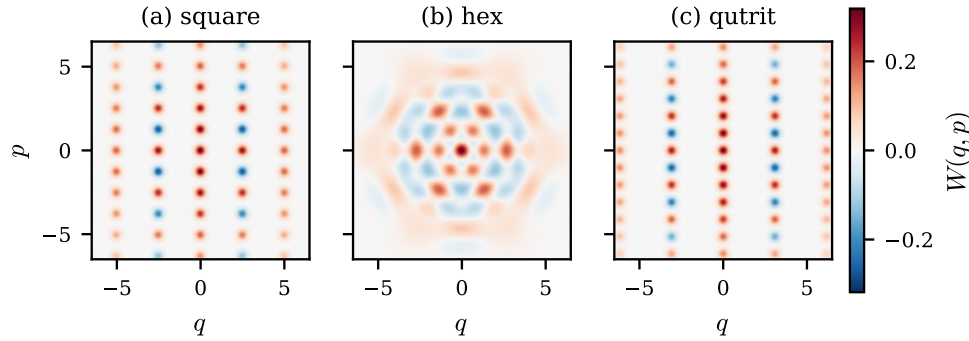


Figure 18. Finite-energy GKP Wigner functions at $\Delta = 0.2$ for three constructions of Chapter 1. *Left:* the square-lattice qubit used throughout the thesis, displayed in the $|0_L\rangle$ coset. *Middle:* the hexagonal-lattice qubit, whose tighter circle packing enlarges the smallest uncorrectable shift by a factor of $(2/\sqrt{3})^{1/2} \approx 1.07457$ [1]. The hex panel displays the full hex stabilizer lattice, equivalently the $|+_L\rangle$ state, because the index-2 $|0_L\rangle$ coset of a triangular lattice is rectangular and would obscure the hexagonal geometry that the panel exists to illustrate. Its peaks are rendered as coherent-state peaks rather than true $\Delta = 0.2$ squeezed peaks, so they look rounder than in panels (a) and (c). *Right:* the square-lattice qutrit ($d_L = 3$), $|0_L\rangle$ coset, whose stabilizer spacing $\sqrt{2\pi d}$ accommodates three codewords in the same phase-space footprint per the Chapter 1.3 sidenote. Panels (a) and (c) share the square-lattice geometry by construction; the only visible difference between them is the q -stabilizer pitch, $2\sqrt{\pi}$ for the qubit versus $\sqrt{2\pi d}$ for the qutrit, which at $d = 3$ is wider by a factor $\sqrt{3/2} \approx 1.22$.

The hexagonal lattice is the tightest packing in two dimensions. For a codeword pair built on the hexagonal lattice at fixed codespace dimension $d_L = 2$, the shortest vector of the symplectic dual lattice has length $(2/\sqrt{3})^{1/2} (1/\sqrt{n})$, compared with $1/\sqrt{n}$ for the square lattice [1]. The smallest uncorrectable shift is therefore larger by a factor of $(2/\sqrt{3})^{1/2} \approx 1.07457$, and the hexagonal code is

correspondingly more robust against small isotropic displacement errors. Near-optimal recovery makes this geometry operational. The leading loss infidelity is controlled by the closest points of the symplectic dual lattice, and the hexagonal expression changes because that closest-point distance increases [5]. The open question is how that geometric gain propagates into the ES-ZNE threshold and exponent.

The second extension is the codespace dimension itself, and Figure 18 displays all three constructions side-by-side. The sidenote at Section 1.3 already records that for a rectangular GKP lattice with stabilizer spacing $d\sqrt{\pi}$, the quotient index is $[\sqrt{\pi}\mathbb{Z} : d\sqrt{\pi}\mathbb{Z}] = d$, so the code encodes a d -dimensional qudit. The square-lattice qubit is the $d = 2$ special case of a one-parameter family [1]. Brock et al. recently demonstrated GKP qutrits and ququarts beyond break-even on the same Yale cQED platform that anchors Chapter 2, with measured QEC gains $G_3 = 1.82 \pm 0.03$ for the qutrit and $G_4 = 1.87 \pm 0.03$ for the ququart [40]. The experimental question of whether ES-ZNE extends from $d_L = 2$ to $d_L > 2$ is therefore well-posed on a device that already exists.

The Petz recovery of Section 3.4 is derived at $d_L = 2$, and the asymptotic infidelity bound of Section 3.5 carries a factor of $1/d_L$ in its dual-lattice form. The single $\langle \tilde{X} \rangle_{\text{cond}}$ of Chapter 3.6 gives way to a set of Weyl-displacement expectation values, and the power-law ansatz (4.6) does not trivially carry over.

SUBSECTION 5.3

Beyond GKP, and the Monotonicity Premise of ES-ZNE

The power-law fit of Equation (4.6) requires the recovered logical observable $y(\bar{n})$ to approach a single limit L through a monotone tail. Section 3.6 establishes this numerically for $\langle \tilde{X} \rangle_{\text{cond}}$ across the loss depths tested, and Chapter 4 validates the ansatz exactly where the log-log residual stays linear. The Albert-Zheng analysis supplies the two conditions that make such an extrapolation meaningful for GKP, namely a well-defined infinite-energy target and a monotone finite-energy trend for the near-optimal infidelity [5, 18, 26]. This monotonicity is also the point at which GKP departs from every other single-mode bosonic code in common use.

Cat codes provide a counterexample. In the pure-loss channel, the optimal coherent-state amplitude that realises the best cat-code fidelity at fixed loss γ is not $\alpha \rightarrow \infty$ but a finite sweet spot $\alpha_\star(\gamma)$, at which the backaction-induced distortion vanishes to leading order [18]. Beyond $\alpha_\star$ the distortion decays as a damped oscillating function with damping coefficient $(1 - \gamma)\alpha^2$, so the fidelity against mean photon number $\bar{n}_{\text{cat}} = |\alpha|^2$ develops peaks and troughs rather than a clean monotone rise. The corresponding curves at loss rate $\gamma = 0.095$ are plotted in [18] for cat codes at spacings $S = 3$ and $S = 4$, and show an oscillatory envelope. A power-law fit run against that curve would not track a monotone tail toward a single limit.

The binomial code is where the contrast is more subtle. For binomial codes the backaction distortion $z_{\ell\ell}^{\text{bin}}$ does not oscillate with $\bar{n}$ [18]. The binomial fidelity nonetheless exhibits its own local maximum in $\bar{n}$ at fixed spacing S , tracking cat's envelope at small and large $\bar{n}$ with only a smoother approach between

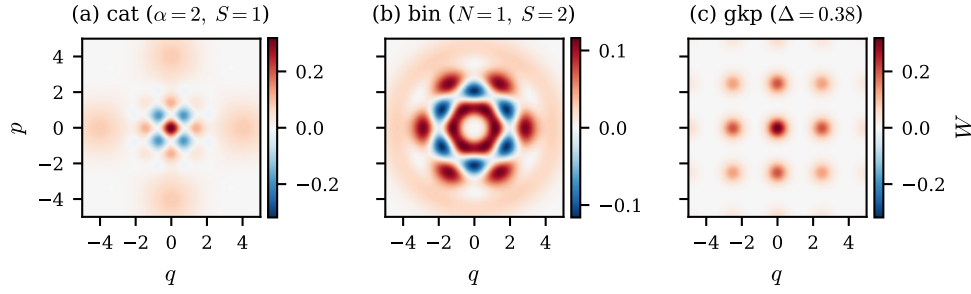


Figure 19. Wigner function of the maximally mixed logical state $\frac{1}{2}(|0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|)$ for three single-mode bosonic codes; color scales are independent. Cat and binomial parameters follow Figure 1 of [18]; the GKP uses $\Delta = 0.38$. *Left:* four-component cat at $\alpha = 2$, $S = 1$ ($\bar{n} \approx 4$), four coherent-state lobes on a circle with interference fringes. *Middle:* binomial at $N = 1$, $S = 2$ ($\bar{n} = 3$), supported on $\{|0\rangle, |3\rangle, |6\rangle\}$ with 3-fold rotational symmetry. *Right:* finite-energy GKP at $\Delta = 0.38$ ($\bar{n} \approx 3$), the alternating-sign lattice of the square code. The three families allocate their phase-space weight qualitatively differently, and each responds differently to the pure-loss channel, as Figure 20 makes quantitative.

them. Along the parametric diagonal $\text{bin}(N, S \approx \xi N)$, sweeping both N and S together, the fidelity does rise monotonically [18]; that trajectory passes through a sequence of distinct binomial codes rather than sweeping $\bar{n}$ within a fixed family, and the local maximum within any one family remains. Figure 19 makes the phase-space contrast among the three families concrete.

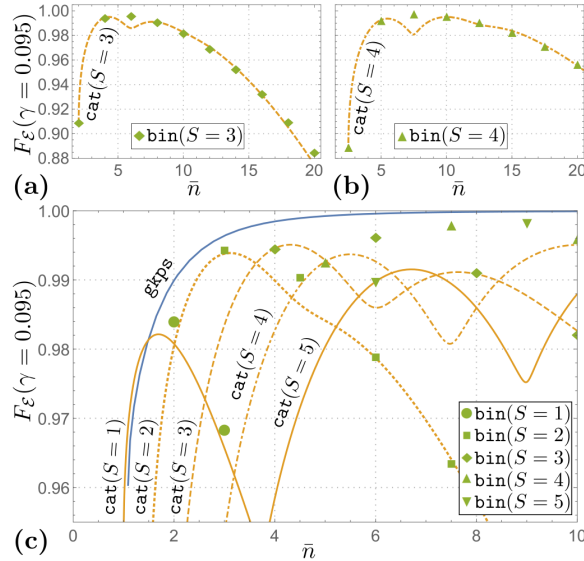


Figure 20. Channel fidelity F_E at loss rate $\gamma = 0.095$ versus mean occupation number $\bar{n}$ for three single-mode bosonic codes, reproduced from [18]. At fixed spacing S , cat codes peak at sweet-spot amplitudes $\alpha_\star(\gamma)$ and develop troughs between them from their oscillatory backaction distortion. Binomial codes are smoother, free of the troughs, but still exhibit a local maximum at finite $\bar{n}$ rather than a monotone rise. Only the GKP curve in panel (c) climbs monotonically with $\bar{n}$, and that monotone tail is the structural feature that makes the Chapter 4 ansatz plausible in the first place.

Among the three single-mode bosonic code families that [18] compares under a common loss-channel and optimal-recovery framework, GKP is therefore the only one whose energy–fidelity curve structurally supports the fit that Chapter 4 runs. Porting ES-ZNE to other codes would require a different protocol, designed around each family’s own energy-fidelity structure. For rotation-symmetric codes, a different mitigation knob is available through the rotation symmetry itself, which supports symmetry expansion and phase-error suppression rather than an energy-axis extrapolation [41]. Figure 20 reproduces Albert et al.’s direct comparison at $\gamma = 0.095$, and the monotone GKP curve in panel (c) is visibly alone among the three families the figure surveys.

Fault-tolerant quantum computation will require logical error rates that current hardware cannot achieve directly. For the recovered logical observable produced by the full GKP loss-recovery-decoding pipeline, a power law is an accurate effective model over the finite-energy window in which the residual diagnostic stays linear, and within that window the fit can be inverted in software to estimate the infinite-energy limit. Whether the same strategy survives into broader noise environments, higher-dimensional codes, and different architectures, remains open.

References

- [1] D. Gottesman, A. Kitaev, and J. Preskill. Encoding a qubit in an oscillator. *Physical Review A*, 2001. doi: 10.1103/PhysRevA.64.012310. URL <http://arxiv.org/pdf/quant-ph/0008040>.
- [2] P. Campagne-Ibarcq, A. Eickbusch, S. Touzard, E. Zalys-Geller, N. E. Frattini, V. V. Sivak, P. Reinhold, S. Puri, S. Shankar, R. J. Schoelkopf, L. Frunzio, M. Mirrahimi, and M. H. Devoret. Quantum error correction of a qubit encoded in grid states of an oscillator. *Nature*, 584:368–372, 2020. doi: 10.1038/s41586-020-2603-3.
- [3] Alexandre Blais, Arne L. Grimsmo, S. M. Girvin, and Andreas Wallraff. Circuit quantum electrodynamics. *Reviews of Modern Physics*, 93(2):025005, 2021. doi: 10.1103/RevModPhys.93.025005.
- [4] Christian Weedbrook, Stefano Pirandola, Raul Garcia-Patrón, Nicolas J. Cerf, Timothy C. Ralph, Jeffrey H. Shapiro, and Seth Lloyd. Gaussian Quantum Information. *Reviews of Modern Physics*, 84:621, 2012. doi: 10.1103/RevModPhys.84.621.
- [5] Guo Zheng, Wenhao He, Gideon Lee, Kyungjoo Noh, and Liang Jiang. Performance and Achievable Rates of the Gottesman-Kitaev-Preskill Code for Pure-Loss and Amplification Channels. *PRX Quantum*, 2025. doi: 10.1103/gh1c-xyn1.
- [6] Jonathan Conrad. *The fabulous world of GKP codes*. PhD Dissertation, Freie Universität Berlin, 2024. URL <https://arxiv.org/abs/2412.02442>.
- [7] Arne L. Grimsmo and Shruti Puri. Quantum Error Correction with the Gottesman-Kitaev-Preskill Code. *PRX Quantum*, 2:020101, 2021. doi: 10.1103/PRXQuantum.2.020101.
- [8] B. Baragiola, Giacomo Pantaleoni, R. N. Alexander, Angela Karanjai, and N. Menicucci. All-Gaussian Universality and Fault Tolerance with the Gottesman-Kitaev-Preskill Code. *Physical Review Letters*, 2019. doi: 10.1103/PhysRevLett.123.200502. URL <https://arxiv.org/pdf/1903.00012>.
- [9] V. V. Sivak, A. Eickbusch, B. Royer, S. Singh, I. Tsioutsios, S. Ganjam, A. Miano, B. L. Brock, A. Z. Ding, L. Frunzio, S. M. Girvin, R. J. Schoelkopf, and M. H. Devoret. Real-time quantum error correction beyond break-even. *Nature*, 2023. doi: 10.1038/s41586-023-05782-6. URL <https://www.nature.com/articles/s41586-023-05782-6>.
- [10] John Clarke and Frank K. Wilhelm. Superconducting quantum bits. *Nature*, 453: 1031–1042, 2008. doi: 10.1038/nature07128.
- [11] Michel H. Devoret, John M. Martinis, and John Clarke. Measurements of macroscopic quantum tunneling out of the zero-voltage state of a current-biased Josephson junction. *Physical Review Letters*, 55(18):1908–1911, 1985. doi: 10.1103/PhysRevLett.55.1908.
- [12] John M. Martinis, Michel H. Devoret, and John Clarke. Energy-level quantization in the zero-voltage state of a current-biased Josephson junction. *Physical Review Letters*, 55(15):1543–1546, 1985. doi: 10.1103/PhysRevLett.55.1543.
- [13] The Nobel Foundation. The Nobel prize in physics 2025. <https://www.nobelprize.org/prizes/physics/2025/summary/>, 2025. Awarded to M. H. Devoret, J. M. Martinis, and J. Clarke for experiments with superconducting circuits that laid the foundation for quantum computing.

- [14] Jens Koch, Terri M. Yu, Jay Gambetta, A. A. Houck, D. I. Schuster, J. Majer, Alexandre Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Charge-insensitive qubit design derived from the cooper pair box. *Physical Review A*, 76(4):042319, 2007. doi: 10.1103/PhysRevA.76.042319.
- [15] Baptiste Royer, Shraddha Singh, and S. M. Girvin. Stabilization of Finite-Energy Gottesman-Kitaev-Preskill States. *Physical Review Letters*, 125:260509, 2020. doi: 10.1103/PhysRevLett.125.260509.
- [16] Alec Eickbusch, Volodymyr Sivak, Andy Z. Ding, Salvatore S. Elder, Shantanu R. Jha, Jayameenakshi Venkatraman, Baptiste Royer, S. M. Girvin, Robert J. Schoelkopf, and Michel H. Devoret. Fast universal control of an oscillator with weak dispersive coupling to a qubit. *Nature Physics*, 18:1464–1469, 2022. doi: 10.1038/s41567-022-01776-9. URL <https://arxiv.org/abs/2111.06414>.
- [17] Anthony J. Brady, A. Eickbusch, Shraddha Singh, Jing Wu, and Quntao Zhuang. Advances in bosonic quantum error correction with Gottesman-Kitaev-Preskill Codes: Theory, engineering and applications. *Progress in Quantum Electronics*, 2023. doi: 10.1016/j.pquantelec.2023.100496. URL <https://arxiv.org/pdf/2308.02913>.
- [18] Victor V. Albert, Kyungjoo Noh, K. Duivenvoorden, Dylan J. Young, R. T. Brierley, P. Reinhold, Christophe Vuillot, Linshu Li, Chao Shen, S. Girvin, B. Terhal, and Liang Jiang. Performance and structure of single-mode bosonic codes. *Physical Review A*, 2017. doi: 10.1103/PhysRevA.97.032346. URL <https://link.aps.org/accepted/10.1103/PhysRevA.97.032346>.
- [19] J. Solomon Ivan, Krishna Kumar Sabapathy, and R. Simon. Operator-sum representation for bosonic Gaussian channels. *Physical Review A*, 84:042311, 2011. doi: 10.1103/PhysRevA.84.042311.
- [20] Per-Olov Löwdin. On the Non-Orthogonality Problem Connected with the Use of Atomic Wave Functions in the Theory of Molecules and Crystals. *Journal of Chemical Physics*, 18:365–375, 1950. doi: 10.1063/1.1747632.
- [21] Mahnaz Jafarzadeh, Jonathan Conrad, Rafael N. Alexander, and Ben Q. Baragiola. Logical channels in approximate Gottesman-Kitaev-Preskill error correction. *Physical Review*, 2025. doi: 10.1103/c8hk-v1qf. URL <https://arxiv.org/abs/2504.13383>.
- [22] H. Barnum and E. Knill. Reversing quantum dynamics with near-optimal quantum and classical fidelity. *Journal of Mathematical Physics*, 43:2097–2106, 2002. doi: 10.1063/1.1459754.
- [23] Hui Khoon Ng and Prabha Mandayam. Simple approach to approximate quantum error correction based on the transpose channel. *Physical Review A*, 81:062342, 2010. doi: 10.1103/PhysRevA.81.062342.
- [24] Dénes Petz. Sufficiency of Channels Over Von Neumann Algebras. *The Quarterly Journal of Mathematics*, 39(1):97–108, 1988. doi: 10.1093/qmath/39.1.97.
- [25] Bikun Li, Zhaoyou Wang, Guo Zheng, Yat Wong, and Liang Jiang. Optimality Condition for the Petz Map. *Physical Review Letters*, 134:200602, 2025. doi: 10.1103/PhysRevLett.134.200602.
- [26] Guo Zheng, Wenhao He, Gideon Lee, and Liang Jiang. Near-Optimal Performance of Quantum Error Correction Codes. *Physical Review Letters*, 132:250602, 2024. doi: 10.1103/PhysRevLett.132.250602.
- [27] Tom B. Harris, Takaya Matsuura, Ben Q. Baragiola, and Nicolas C. Menicucci.

- Logical channel for heralded and pure loss with the Gottesman-Kitaev-Preskill code. *Physical Review A*, 112:042417, 2025. doi: 10.1103/rmlm-vbfd.
- [28] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error Mitigation for Short-Depth Quantum Circuits. *Physical Review Letters*, 119:180509, 2017. doi: 10.1103/PhysRevLett.119.180509.
- [29] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J. Zeng. Digital zero noise extrapolation for quantum error mitigation. In *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316, 2020. doi: 10.1109/QCE49297.2020.00045.
- [30] Zhenyu Cai, Ryan Babbush, Simon C. Benjamin, Suguru Endo, William J. Huggins, Ying Li, Jarrod R. McClean, and Thomas E. O’Brien. Quantum error mitigation. *Reviews of Modern Physics*, 95:045005, 2023. doi: 10.1103/RevModPhys.95.045005.
- [31] Misty A. Wahl, Andrea Mari, Nathan Shammah, William J. Zeng, and Gokul Subramanian Ravi. Zero noise extrapolation on logical qubits by scaling the error correction code distance, 2023. arXiv:2304.14004.
- [32] Suguru Endo, Keitaro Anai, Yuichiro Matsuzaki, Yuuki Tokunaga, and Yasunari Suzuki. Projective squeezing for translation symmetric bosonic codes, 2025. arXiv preprint.
- [33] A. S. Holevo and R. F. Werner. Evaluating capacities of bosonic Gaussian channels. *Physical Review A*, 63:032312, 2001. doi: 10.1103/PhysRevA.63.032312.
- [34] Jacob Hastrup and Ulrik Lund Andersen. Analysis of loss correction with the Gottesman-Kitaev-Preskill code. *Physical Review A*, 2023. arXiv:2308.01852.
- [35] Ryuji Takagi, Hiroyasu Tajima, and Mile Gu. Universal Sampling Lower Bounds for Quantum Error Mitigation. *Physical Review Letters*, 131(21):210602, 2023. doi: 10.1103/PhysRevLett.131.210602.
- [36] Ryuji Takagi, Suguru Endo, Shintaro Minagawa, and Mile Gu. Fundamental limits of quantum error mitigation. *npj Quantum Information*, 8:114, 2022. doi: 10.1038/s41534-022-00618-z.
- [37] Francesco Mezzadri. How to generate random matrices from the classical compact groups. *Notices of the American Mathematical Society*, 54:592–604, 2007.
- [38] Peter Leviant, Qian Xu, Liang Jiang, and Serge Rosenblum. Quantum capacity and codes for the bosonic loss-dephasing channel. *Quantum*, 6:821, 2022. doi: 10.22331/q-2022-09-29-821. URL <https://quantum-journal.org/papers/q-2022-09-29-821/>.
- [39] Mackenzie H. Shaw, Andrew C. Doherty, and Arne L. Grimsmo. Stabilizer Subsystem Decompositions for Single- and Multimode Gottesman-Kitaev-Preskill Codes. *PRX Quantum*, 5:010331, 2024. doi: 10.1103/PRXQuantum.5.010331.
- [40] Benjamin L. Brock, Shraddha Singh, Alec Eickbusch, Volodymyr V. Sivak, Andy Z. Ding, Luigi Frunzio, Steven M. Girvin, and Michel H. Devoret. Quantum error correction of qudits beyond break-even. *Nature*, 2025. doi: 10.1038/s41586-025-08899-y.
- [41] Suguru Endo, Yasunari Suzuki, Kento Tsubouchi, Rui Asaoka, Kaoru Yamamoto, Yuichiro Matsuzaki, and Yuuki Tokunaga. Quantum error mitigation for rotation-symmetric bosonic codes with symmetry expansion. *Physical Review A*, 111:062402, 2025. doi: 10.1103/PhysRevA.111.062402.